

LO Learning Objectives

1. Understand the physical significance of frequency domain analysis, and compute the frequency response of circuits using AC circuit analysis tools. *Section 6.1.*
2. Compute the Fourier spectrum of periodic signals by using the Fourier series representation, and use this representation in connection with frequency response ideas to compute the response of circuits to periodic inputs. *Section 6.2.*
3. Analyze simple first- and second-order electrical filters, and determine their frequency response and filtering properties. *Section 6.3.*
4. Compute the frequency response of a circuit and its graphical representation in the form of a Bode plot. *Section 6.4.*

6.1 SINUSOIDAL FREQUENCY RESPONSE

The **sinusoidal frequency response** (or, simply, **frequency response**) of a circuit provides a measure of how the circuit responds to sinusoidal inputs of arbitrary frequency. In other words, for a given input signal with a particular amplitude, phase, and frequency, the frequency response of a circuit permits the computation of a particular output signal. For example, suppose you wanted to determine how the load voltage V_o or current I_o varied in response to different frequencies in the circuit of Figure 6.1. An analogy could be made, for example, with how an earphone (the load) responds to the audio signal generated by an MP3 player (the source) when an amplifier (the circuit) is placed between the two.¹ In the circuit of Figure 6.1, the signal source circuitry is represented by a Thévenin source. The impedances are, in general, a function of frequency. The amplifier circuit is represented by the idealized connection of two impedances Z_1 and Z_2 , and the load is represented by an additional impedance Z_o . The following statement provides a general definition of the frequency response of such a system:



The frequency response of a circuit is a measure of the variation of a load-related voltage or current as a function of the frequency of the excitation signal.

Frequency Response Functions

A frequency response function is the ratio of a chosen *output* to a chosen *input*. In circuit analysis, the chosen input is often an independent voltage or current source. The chosen output can be any voltage or current elsewhere in the circuit. By convention, frequency response functions are represented by either **G** or **H**, where **G** is a

¹The circuitry in a high-fidelity audio system is far more complex than the circuits discussed in this chapter. However, from the standpoint of intuition and everyday experience, the audio analogy provides a useful example. The audio spectrum terms *bass*, *midrange*, and *treble* are well known, but not well understood. The material presented in this chapter provides a technical basis for understanding these concepts.

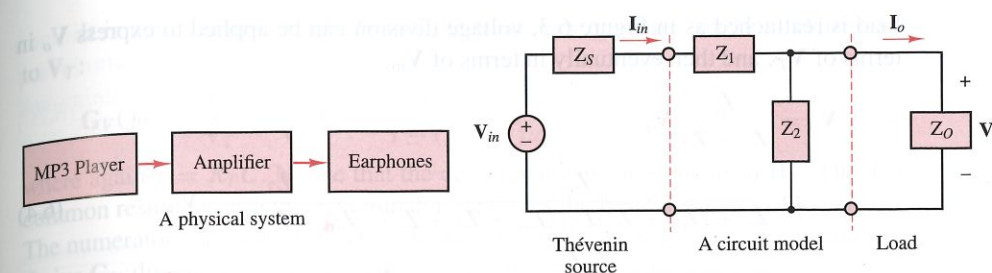


Figure 6.1 A circuit model

dimensionless *gain* and **H** is either an impedance or conductance. Four distinct types of frequency response function are defined as follows:

$$\begin{aligned} G_V(j\omega) &= \frac{V_o(j\omega)}{V_{in}(j\omega)} & G_I(j\omega) &= \frac{I_o(j\omega)}{I_{in}(j\omega)} \\ H_Z(j\omega) &= \frac{V_o(j\omega)}{I_{in}(j\omega)} & H_Y(j\omega) &= \frac{I_o(j\omega)}{V_{in}(j\omega)} \end{aligned} \quad (6.1)$$



In many cases the inputs V_{in} and I_{in} are chosen to be independent voltage and current sources, respectively. The outputs V_o and I_o are freely chosen and, as such, represent the load in a circuit.

The above frequency response functions are related. For example, if $G_V(j\omega)$ and $G_I(j\omega)$ are known, the other two can be derived directly:

$$H_Z(j\omega) = \frac{V_o(j\omega)}{I_{in}(j\omega)} = \frac{I_o(j\omega)}{I_{in}(j\omega)} Z_o(j\omega) = G_I(j\omega) Z_o(j\omega) \quad (6.2)$$

$$H_Y(j\omega) = \frac{I_o(j\omega)}{V_{in}(j\omega)} = \frac{V_o(j\omega)}{Z_o(j\omega)} \frac{1}{V_{in}(j\omega)} = \frac{G_V(j\omega)}{Z_o(j\omega)} \quad (6.3)$$

Frequency response functions are important because they express the frequency response as a single function that relates an output (load) voltage or current to a given input.

Circuit Simplification

In general, the first step in determining the details of a chosen frequency response function is to divide the circuit into a load (in accord with the chosen output) and a source. Consider again the circuit shown in Figure 6.1. The network attached to the load can be replaced by its Thévenin equivalent as shown in Figure 6.2. Once the

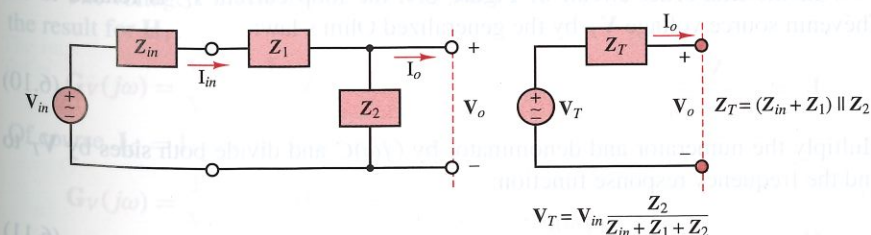


Figure 6.2 Thévenin equivalent source network

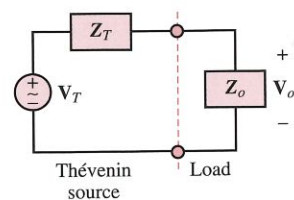


Figure 6.3 Equivalent circuit from the perspective of the load

load is reattached as in Figure 6.3, voltage division can be applied to express V_o in terms of V_T , and then eventually in terms of V_{in} .

$$\begin{aligned} V_o &= \frac{Z_o}{Z_o + Z_T} V_T \\ &= \frac{Z_o}{Z_o + (Z_{in} + Z_1)Z_2 / (Z_{in} + Z_1 + Z_2)} \cdot \frac{Z_2}{Z_{in} + Z_1 + Z_2} V_{in} \\ &= \frac{Z_o Z_2}{Z_o(Z_{in} + Z_1 + Z_2) + (Z_{in} + Z_1)Z_2} V_{in} \end{aligned} \quad (6.4)$$

The gain, $G_V(j\omega)$, is a dimensionless complex quantity, given by:

$$G_V(j\omega) = \frac{V_o}{V_{in}}(j\omega) = \frac{Z_o Z_2}{Z_o(Z_{in} + Z_1 + Z_2) + (Z_{in} + Z_1)Z_2} \quad (6.5)$$

Thus, the gain is known if the circuit element impedances are known.



$V_o(j\omega)$ is a phase-shifted and amplitude-scaled version of $V_{in}(j\omega)$.

If the phasor source voltage and the frequency response of the circuit are known, the phasor load voltage can be computed as follows:

$$V_o(j\omega) = G_V(j\omega) \cdot V_{in}(j\omega) \quad (6.6)$$

$$V_o e^{j\phi_o} = |G_V| e^{j\angle G_V} \cdot V_{in} e^{j\phi_{in}} \quad (6.7)$$

such that

$$V_o = |G_V| \cdot V_{in} \quad (6.8)$$

and

$$\phi_o = \angle G_V + \phi_{in} \quad (6.9)$$

At any given angular frequency ω , the load voltage is a sinusoid with the same frequency as the source voltage.

First- and Second-Order Archetypes

Whenever possible, the first step toward deriving a frequency response function is to use Thévenin's or Norton's theorem to simplify the circuit. If the circuit is first-order, or second-order with the storage elements in series or parallel, it can be simplified to one of the four archetypes shown in Figures 6.4 to 6.7.

In the first-order circuit of Figure 6.4, the loop current I_C is related to the Thévenin source voltage V_T by the generalized Ohm's law:

$$I_C = \frac{V_T}{R_T + Z_C} \quad (6.10)$$

Multiply the numerator and denominator by $(j\omega)C$ and divide both sides by V_T to find the frequency response function:

$$H_Y(j\omega) = \frac{I_C}{V_T} = \frac{(j\omega)C}{1 + (j\omega)\tau} \quad (6.11)$$

where $\tau = R_T C$.

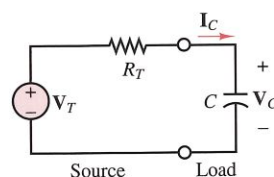


Figure 6.4 Simplified first-order circuit with one capacitor

It is now a simple matter to find the frequency response function relating V_C to V_T :

$$G_V(j\omega) = \frac{V_C}{V_T} = \frac{I_C}{V_T} Z_C = \frac{(j\omega)C}{1 + (j\omega)\tau} \frac{1}{(j\omega)C} = \frac{1}{1 + (j\omega)\tau} \quad (6.12)$$

where again $\tau = R_T C$. Notice that the denominator is the same as in H_Y . This is a common result because the denominator expresses the basic dynamics of the circuit. The numerator expresses differences in the circuit variables. It is a useful exercise to derive G_V directly from voltage division. Try it!

A similar approach can be taken to find the frequency response relating the voltage V_L to the Norton source current I_N in the first-order circuit of Figure 6.5. Apply the generalized Ohm's law to write:

$$V_L = I_N (R_N \parallel Z_L) = I_N \frac{R_N(j\omega)L}{R_N + (j\omega)L} \quad (6.13)$$

The frequency response function is found by dividing both sides by I_N and then dividing the numerator and denominator by R_N .

$$H_Z(j\omega) = \frac{V_L}{I_N} = \frac{(j\omega)L}{1 + (j\omega)\tau} \quad (6.14)$$

where, in this case, $\tau = L/R_N$.

Again, it is a simple matter to find the frequency response function relating I_L to I_N :

$$G_I(j\omega) = \frac{I_L}{I_N} = \frac{V_L}{I_N Z_L} = \frac{1}{1 + (j\omega)\tau} \quad (6.15)$$

where again $\tau = L/R_N$. Also, once more the denominator is the same as in H_Z . It is a useful exercise to derive G_I directly from current division. Try it!

Second-order circuits are handled in much the same way. Consider the series LC circuit of Figure 6.6. The common loop current I_L is related to the Thévenin source voltage V_T by the generalized Ohm's law:

$$V_T = I_L (R_T + Z_C + Z_L) \quad (6.16)$$

Divide both sides by I_L , invert both sides, and multiply the resulting numerator and denominator by $j\omega C$ to find:

$$\begin{aligned} H_Y(j\omega) &= \frac{I_L}{V_T} = \frac{(j\omega)C}{1 + (j\omega)R_T C + (j\omega)^2 LC} \\ &= \frac{(j\omega)C}{1 + (j\omega)\tau + (j\omega/\omega_n)^2} \end{aligned} \quad (6.17)$$

where $\tau = R_T C$ and $\omega_n^2 = 1/LC$. This latter term is the same natural frequency often found in second-order transient circuits.

The voltage gain G_V for the second-order series LC circuit can be found using the result for H_Y .

$$G_V(j\omega) = \frac{V_C}{V_T} = \frac{I_C}{V_T} Z_C \quad (6.18)$$

Of course, $I_C = I_L$ for the series loop, so:

$$G_V(j\omega) = \frac{I_L}{V_T} Z_C = H_Y Z_C = \frac{1}{1 + (j\omega)\tau + (j\omega/\omega_n)^2} \quad (6.19)$$

where again $\tau = R_T C$ and $\omega_n^2 = 1/LC$.

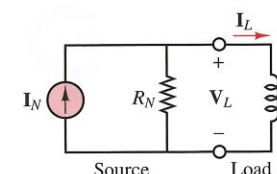


Figure 6.5 Simplified first-order circuit with one inductor

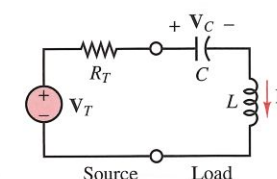


Figure 6.6 Simplified second-order circuit with one capacitor and one inductor in series

Finally, Figure 6.7 shows a second-order parallel LC circuit. The common voltage V_C is related to the Norton source current I_N by the generalized Ohm's law:

$$V_C = I_N (R_N \parallel Z_L \parallel Z_C) \quad (6.20)$$

Divide both sides by I_N to obtain:

$$H_Z(j\omega) = \frac{V_C}{I_N} = \frac{1}{1/R_N + 1/j\omega L + j\omega C} \quad (6.21)$$

Multiply the numerator and denominator by $j\omega L$ to obtain:

$$H_Z(j\omega) = \frac{(j\omega)L}{1 + (j\omega)L/R_N + (j\omega)^2 LC} = \frac{(j\omega)L}{1 + (j\omega)\tau + (j\omega/\omega_n)^2} \quad (6.22)$$

where $\tau = L/R_N$ and $\omega_n^2 = 1/LC$ is the same natural frequency often found in the second-order series LC circuit.

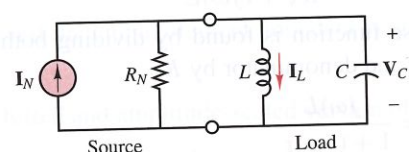


Figure 6.7 Simplified second-order circuit with one capacitor and one inductor in parallel

The current gain G_I for the second-order parallel LC circuit can be found using the result for H_Z .

$$G_I(j\omega) = \frac{I_L}{I_N} = \frac{V_L}{I_N Z_L} \quad (6.23)$$

Of course, $V_L = V_C$, so:

$$G_I(j\omega) = \frac{V_L}{I_N Z_L} = \frac{H_Z}{Z_L} = \frac{1}{1 + (j\omega)\tau + (j\omega/\omega_n)^2} \quad (6.24)$$

where again $\tau = L/R_N$ and $\omega_n^2 = 1/LC$.

Poles and Zeros

By definition, a frequency response function is the ratio of an output to an input. Consequently, the development of any specific frequency response function will, in general, also result in a ratio. The numerator and denominator of any frequency response function can always be expressed as the product of four distinct terms. One of these terms is simply a constant. The other three terms are known as *zeros* or *poles*, depending upon whether they appear in the numerator or denominator, respectively. Each term has its own name, as indicated in the following list.

1. K A constant
2. $(j\omega)$ Pole or zero at the origin
3. $(1 + j\omega\tau)$ Simple pole or zero
4. $[1 + j\omega\tau + (j\omega/\omega_n)^2]$ Quadratic (complex) pole or zero

A simple pole or zero may also take the form $(1 + j\omega/\omega_0)$, where $\omega_0 = 1/\tau$.

The first- and second-order frequency response functions developed in the previous section are good examples of the standard form in which the numerator and denominator are expressed as products of these four terms. These same terms will appear repeatedly later in this chapter when filters and Bode plots are discussed.

EXAMPLE 6.1 Computing the Frequency Response Using Thévenin's Theorem

Problem

Compute the frequency response $G_V(j\omega) = V_o/V_s$ for Figure 6.8.

Solution

Known Quantities: $R_1 = 10 \text{ k}\Omega$; $C = 10 \text{ }\mu\text{F}$; $R_o = 10 \text{ k}\Omega$.

Find: The frequency response $G_V(j\omega) = V_o/V_s$.

Assumptions: None.

Analysis: With R_o as the load resistance, the approach is to use Thévenin's theorem to determine the equivalent network of the source network; that is, the equivalent network of everything to the left of terminals a and b . The Thévenin equivalent impedance Z_T of the source network is:

$$Z_T = (R_1 \parallel Z_C)$$

The Thévenin (open-circuit) voltage V_T across terminals a and b is found from voltage division:

$$V_T = V_s \frac{Z_C}{R_1 + Z_C}$$

After reattaching the load to the Thévenin source, the voltage V_o across the load can be found by applying voltage division once more:

$$\begin{aligned} V_o &= \frac{R_o}{Z_T + R_o} V_T \\ &= \frac{R_o}{R_1 Z_C / (R_1 + Z_C) + R_o} \frac{Z_C}{R_1 + Z_C} V_s \end{aligned}$$

Thus:

$$G_V(j\omega) = \frac{V_o}{V_s}(j\omega) = \frac{R_o Z_C}{R_o(R_1 + Z_C) + R_1 Z_C}$$

The impedances of the circuit elements are $R_1 = 10^3 \Omega$, $Z_C = 1/(j\omega \times 10^{-5}) \Omega$, and $R_o = 10^4 \Omega$. The resulting frequency response is:

$$\begin{aligned} G_V(j\omega) &= \frac{10^4}{10^4 \left(10^3 + \frac{1}{j\omega \times 10^{-5}} \right) + \frac{10^3}{j\omega \times 10^{-5}}} = \frac{100}{110 + j\omega} \\ &= \frac{100}{\sqrt{110^2 + \omega^2}} \angle -\tan^{-1}(\omega/110) \end{aligned}$$

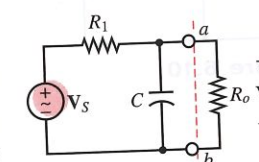


Figure 6.8

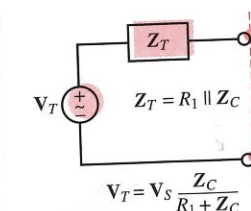


Figure 6.9

Comments: The use of equivalent circuit ideas is often helpful in deriving frequency response functions because it naturally forces us to identify source and load quantities. However, it is certainly not the only method of solution. For example, node analysis would have yielded the same results just as easily, by recognizing that the top node voltage is equal to the load voltage and by solving directly for V_o as a function of V_s , without going through the intermediate step of computing the Thévenin equivalent source circuit.

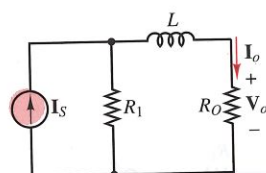


Figure 6.10

EXAMPLE 6.2 Computing the Frequency Response

Problem

Compute the frequency response $H_Z(j\omega) = V_o/I_s$ for Figure 6.10.

Solution

Known Quantities: $R_1 = 1 \text{ k}\Omega$; $L = 2 \text{ mH}$; $R_o = 4 \text{ k}\Omega$.

Find: The frequency response $H_Z(j\omega) = V_o/I_s$.

Assumptions: None.

Analysis: While it is possible to find the Thévenin or Norton equivalent network of everything attached to R_o and proceed as in the previous example to find the frequency response function, it is also possible to apply current division to find I_o and then apply Ohm's law to find V_o and thus the frequency response function.

Apply current division to write:

$$I_o = \frac{R_1}{R_1 + R_o + j\omega L} I_s$$

Factor out $R_1 + R_o$ in the denominator to find:

$$I_o = \frac{R_1}{R_1 + R_o} \frac{1}{1 + j\omega L/(R_1 + R_o)} I_s$$

Then:

$$\begin{aligned} \frac{V_o}{I_s}(j\omega) &= H_Z(j\omega) = \frac{I_o R_o}{I_s} \\ &= \frac{R_1 R_o}{R_1 + R_o} \frac{1}{1 + j\omega L/(R_1 + R_o)} \end{aligned}$$

Substitute numerical values to obtain:

$$\begin{aligned} H_Z(j\omega) &= \frac{(10^3)(4 \times 10^3)}{10^3 + 4 \times 10^3} \frac{1}{1 + (j\omega)(2 \times 10^{-3})/(5 \times 10^3)} \\ &= (0.8 \times 10^3) \frac{1}{1 + j0.4 \times 10^{-6} \omega} \end{aligned}$$

Comments: The units of $H_Z(j\omega)$ should be ohms. Verify that they are!

CHECK YOUR UNDERSTANDING

Refer to Example 6.1 and compute the magnitude and phase of G_V at the frequencies $\omega = 10$, 100, and 1,000 rad/s.

Answers: Magnitude = 0.9054, 0.6727, and 0.0994; phase (degrees) = -5.1944, -42.2737, and -83.7227

CHECK YOUR UNDERSTANDING

Refer to Example 6.2 and compute the magnitude and phase of H_Z at the frequencies $\omega = 1$, 10, and 100 Mrad/s.

Answer: Magnitude = 742.78 Ω , 194.03 Ω , and 19.99 Ω ; phase (degrees) = -21.8°, -75.96°, and -88.57°

6.2 FOURIER ANALYSIS

The aim of this section is to introduce the concept of frequency domain analysis of signals, and specifically the **Fourier series**. In the next few pages we explain how it is possible to represent periodic signals by means of the superposition of various sinusoidal signals of different amplitude, phase, and frequency. Let the signal $x(t)$ be periodic with period T , that is,

$$x(t) = x(t + T) = x(t + nT) \quad n = 1, 2, 3, \dots; T = \text{period} \quad (6.25)$$

An example of a periodic signal is shown in Figure 6.11.

The signal $x(t)$ can be expressed as an infinite summation of sinusoidal components, known as a **Fourier series**, using either of the following two representations.

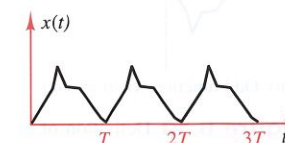


Figure 6.11 A periodic signal

Fourier Series

1. Sine-cosine (quadrature) representation

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(n \frac{2\pi}{T} t\right) + \sum_{n=1}^{\infty} b_n \sin\left(n \frac{2\pi}{T} t\right) \quad (6.26)$$

2. Magnitude and phase form

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \sin\left(n \frac{2\pi}{T} t + \theta_n\right) \quad (6.27)$$

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos\left(n \frac{2\pi}{T} t - \psi_n\right) \quad (6.28)$$

In each of these expressions, the period T is related to the **fundamental frequency** of the signal ω_0 by

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T} \quad \text{rad/s} \quad (6.29)$$

the frequencies $2\omega_0$, $3\omega_0$, $4\omega_0$, etc., are called **harmonics**.



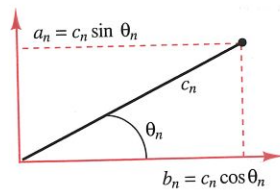
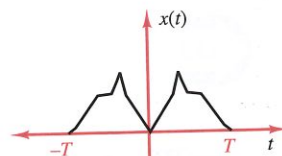
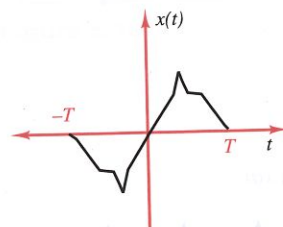


Figure 6.12 Relationship between $\{a_n, b_n\}$ and $\{c_n, \theta_n\}$ forms



(a) Even function, $x(-t) = x(t)$



(b) Odd function, $x(-t) = -x(t)$

Figure 6.13 Definition of even and odd functions

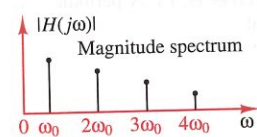


Figure 6.14 Discrete frequency spectrum

It is straightforward to show that equations 6.26 and 6.27 are equivalent when the parameters a_n , b_n , c_n , and θ_n are defined by:

$$\sqrt{a_n^2 + b_n^2} = c_n \quad \text{and} \quad \frac{b_n}{a_n} = \cot(\theta_n) \quad (6.30)$$

Similarly, one can show that equations 6.26 and 6.28 are equivalent when the parameters a_n , b_n , c_n , and ψ_n are defined by:

$$\sqrt{a_n^2 + b_n^2} = c_n \quad \text{and} \quad \frac{b_n}{a_n} = \tan(\psi_n) \quad (6.31)$$

Figure 6.12 is a graphical representation of the equivalence of the $\{a_n, b_n\}$ and $\{c_n, \theta_n\}$ forms of the Fourier series.

Each of the two forms of the Fourier series, equations 6.26 and 6.27 (or 6.28), has its distinct advantages. The sine-cosine representation uses odd and even functions of the independent variable. Odd functions are antisymmetric about the origin and satisfy the condition:

$$f(-t) = -f(t) \quad \text{odd function} \quad (6.32)$$

Sine functions are odd. Even functions are symmetric about the origin and satisfy the condition:

$$f(-t) = f(t) \quad \text{even function} \quad (6.33)$$

Cosine functions are even, as is any constant, such as a_0 . Figure 6.13 shows even and odd function examples.

The advantage of the representation in equation 6.26 is that if $x(t)$ is known to be odd (even), it can be represented as the sum of only odd (even) functions [i.e., using only the sine (cosine) terms], thus resulting in easier evaluation of the Fourier series coefficients.

The magnitude and phase forms, equations 6.27 and 6.28, separate out the magnitude information c_n from the phase information θ_n or ψ_n . In this form, Fourier series may be combined readily with magnitude and phase representations of linear systems to periodic inputs, which are described in the following section. The magnitude and phase components are often represented as a discrete **frequency spectrum**, as shown in Figure 6.14.

Computation of Fourier Series Coefficients

The computation of the $\{a_n, b_n\}$ or $\{c_n, \theta_n\}$ coefficients for the periodic function $x(t)$ is based on the following formulas:

$$a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt = \text{average value of } x(t) \quad (6.34)$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos\left(n \frac{2\pi}{T} t\right) dt = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos\left(n \frac{2\pi}{T} t\right) dt \quad (6.35)$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin\left(n \frac{2\pi}{T} t\right) dt = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin\left(n \frac{2\pi}{T} t\right) dt \quad (6.36)$$

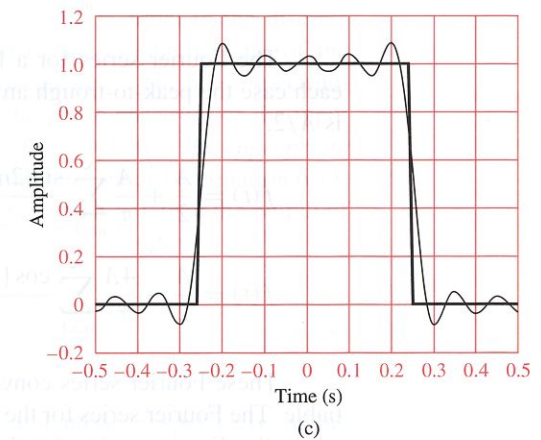
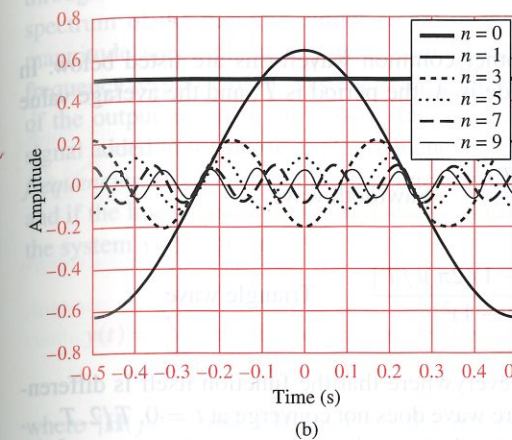
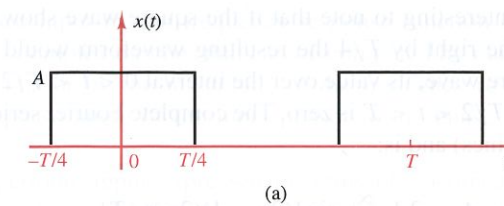


Figure 6.15 Square wave and its representation by a Fourier series. (a) Square wave (even function); (b) first six Fourier series terms of square wave; (c) sum of first six Fourier series terms superimposed upon a square wave

The integral limits are written in two different forms to illustrate that it does not matter where the integration starts, provided that it is carried out over one entire period. The c_n and θ_n (or ψ_n) values can be derived from the a_n and b_n coefficients by using equations 6.30 and 6.31.

To illustrate the significance of the Fourier series decomposition, consider the square wave of Figure 6.15(a), which is an even function. Only even function (cosine) terms are non-zero. The first six non-zero Fourier series terms are shown in Figure 6.15(b). Note that the first term corresponds to a_0 , which is the average value $A/2$ of the function. The other five terms correspond to n odd (1, 3, 5, 7, and 9). Note also that the coefficients for $n = 1, 5$, and 9 are positive, and those for $n = 3$ and 7 are negative (you can see this by looking at the peaks of the cosine waveforms). This alternation of positive and negative cosines is required so that each term can add or subtract from the previous terms as needed to “flatten” the waveform into a square wave. Figure 6.15(c) compares the original square wave with the Fourier series approximation. It should be evident that 10 terms are not sufficient to reproduce the sharp edges of the square wave, but it should also be clear that as we add more Fourier terms, the resulting approximation will be closer and closer to the square wave signal.

The complete Fourier series for the square wave shown in Figure 6.15(a) is:

$$f(t) = \frac{A}{2} + \frac{2A}{\pi} \sum_{n=1}^{\infty} (-1)^{(n-1)} \frac{\cos[(2n-1)(2\pi)t/T]}{2n-1}$$

It is interesting to note that if the square wave shown in Figure 6.15(a) were shifted to the right by $T/4$ the resulting waveform would be an odd function. For such a square wave, its value over the interval $0 < t < T/2$ is A while its value over the interval $T/2 < t < T$ is zero. The complete Fourier series now contains only odd functions (sines) and is:

$$f(t) = \frac{A}{2} + \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{\sin[(2n-1)(2\pi)t/T]}{2n-1}$$

The Fourier series for a few other common waveforms are listed below. In each case the peak-to-trough amplitude is A , the period is T , and the average value is $A/2$.

$$f(t) = \frac{A}{2} + \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n\pi t/T)}{n} \quad \text{Sawtooth wave}$$

$$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos[(2n-1)(2\pi)t/T]}{(2n-1)^2} \quad \text{Triangle wave}$$

These Fourier series converge everywhere that the function itself is differentiable. The Fourier series for the square wave does not converge at $t = 0, T/2, T, \dots$ while the Fourier series for the sawtooth wave does not converge at $t = 0, T, 2T, \dots$

Response of Linear Systems to Periodic Inputs

The frequency response concept is particularly useful when one deals with a system excited by a periodic input, which can be modeled by a Fourier series of sinusoids of known amplitude and phase, but different frequencies. Assume that the Fourier series has a finite number of terms:

$$x(t) = c_0 + \sum_{n=1}^N c_n \sin\left(n \frac{2\pi}{T} t + \theta_n\right) \quad (6.37)$$

Each of the N sinusoids is characterized by amplitude c_n , phase θ_n , and frequency $\omega_n = n\omega_0$, where $\omega_0 = 2\pi/T$ and T is the period of the input signal. For example, the periodic input could be the sawtooth waveform shown in Example 6.3.

Figure 6.16 illustrates the input-output representation of a system, making use of the frequency response concept. The figure shows that if the input $q_{in}(t)$ to a linear system can be represented by the phasor $Q_{in}(j\omega)$, then the output can be computed

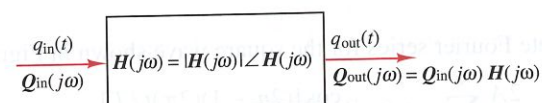


Figure 6.16 Response of a linear system to a phasor input

by multiplying the phasor input by the frequency response function of the linear system. This product is a complex number that can be computed by multiplying the magnitude of the input phasor by the magnitude of the frequency response function and by adding the phase angle of the input phasor to the phase angle of the frequency response.

In the case of a periodic input expressed in terms of a (truncated) Fourier series, we must recognize that each of the input sinusoidal components propagates through the system according to the frequency response. Thus, the discrete magnitude spectrum of the periodic output signal in the steady state is equal to the discrete magnitude spectrum of the input signal multiplied by the amplitude ratio of the frequency response of the system at the appropriate frequencies. The phase spectrum of the output signal in the steady state is equal to the phase spectrum of the input signal added to the phase angle frequency response of the system at the appropriate frequencies. If $x(t)$ is the input to a linear system in the form given by equation 6.13, and if the linear system has a frequency response function $H(j\omega)$, then the output of the system $y(t)$ is given by

$$y(t) = \sum_{n=1}^N |H(j\omega_n)| c_n \sin[\omega_n t + \theta_n + \angle H(j\omega_n)] \quad (6.38)$$

where $|H(j\omega_n)|$ and $\angle H(j\omega_n)$ are the magnitude and phase, respectively, at the frequency corresponding to the n th harmonic $n\omega_0$ of the input.

EXAMPLE 6.3 Computation of Fourier Series Coefficients

Problem

Compute the complete Fourier spectrum of the sawtooth function shown in Figure 6.17; that is, find a general expression for the coefficients a_n and b_n as a function of n , and then compute the spectrum of $x(t)$, that is, the coefficients c_n and θ_n . Plot the spectrum of the signal.

Solution

Known Quantities: Amplitude and period of sawtooth waveform.

Find: Fourier series coefficients a_n and b_n .

Schematics, Diagrams, Circuits, and Given Data: The function is periodic, with period $T = 1$ s and peak amplitude $A = 1$.

Assumptions: None.

Analysis: The function in Figure 6.17(a) is odd since it is antisymmetric about the vertical axis. Thus, only the b_n coefficients are non-zero. First, express $x(t)$ over one period:

$$x(t) = A \left(1 - \frac{2t}{T}\right) \quad 0 \leq t < T$$

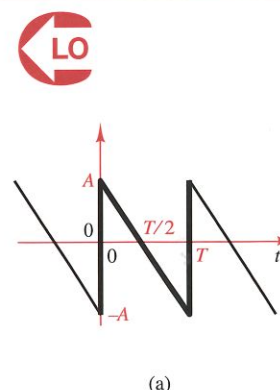


Figure 6.17 (a) Periodic (sawtooth) function

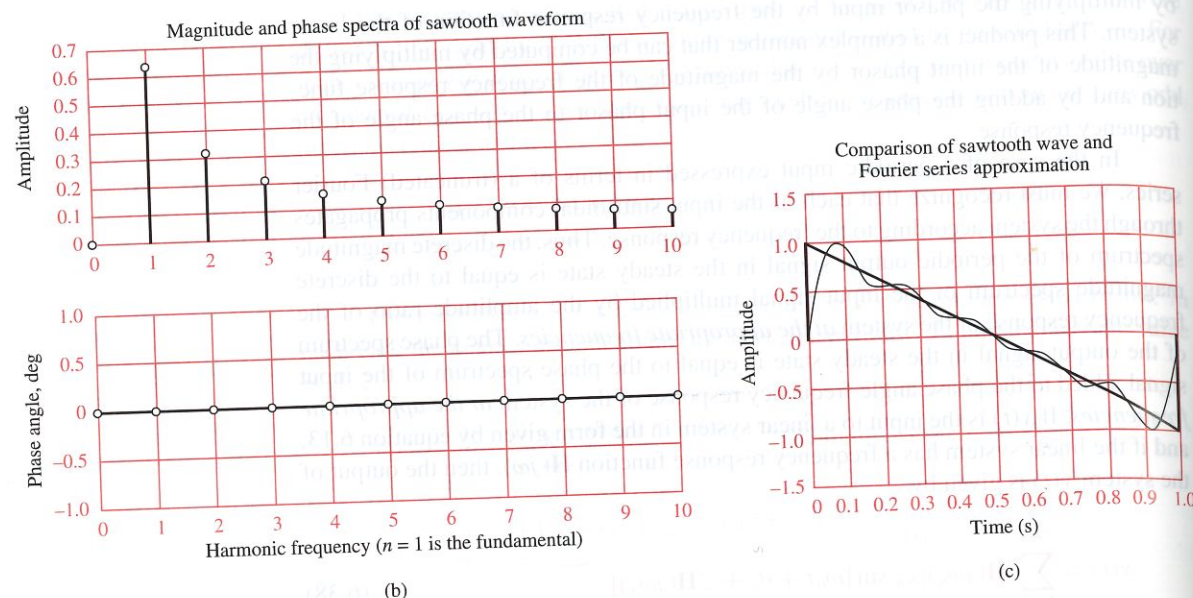


Figure 6.17 (b) spectrum of sawtooth waveform; (c) approximation of sawtooth waveform for $N = 5$

Then we evaluate the integral in equation 6.36:

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T A \left(1 - \frac{2t}{T}\right) \sin\left(n \frac{2\pi}{T} t\right) dt \\
 &= \frac{2}{T} \int_0^T A \sin\left(n \frac{2\pi}{T} t\right) dt + \frac{2A}{T} \int_0^T \left(-\frac{2t}{T}\right) \sin\left(n \frac{2\pi}{T} t\right) dt \\
 &= \frac{2A}{T} \left[-\frac{T}{2n\pi} \cos\left(n \frac{2\pi}{T} t\right) \right]_0^T - \frac{4A}{T^2} \int_0^T t \cdot \sin\left(n \frac{2\pi}{T} t\right) dt \\
 &= 0 - \frac{4A}{T^2} \left[\frac{1}{n^2(2\pi/T)^2} \sin\left(n \frac{2\pi}{T} t\right) - \frac{t}{n(2\pi/T)} \cos\left(n \frac{2\pi}{T} t\right) \right]_0^T \\
 &= -\frac{4A}{T^2} \left[-\frac{T^2}{2n\pi} \cos(2n\pi) \right] = \frac{2A}{n\pi} \quad n = 1, 2, 3, \dots
 \end{aligned}$$

To compute the spectrum of the signal, apply equation 6.30:

$$\begin{aligned}
 c_n &= \sqrt{a_n^2 + b_n^2} = |b_n| \\
 \theta_n &= \cot^{-1} \frac{b_n}{a_n} = \cot^{-1} \frac{b_n}{0} = 0
 \end{aligned}$$

The individual components of the spectrum of $x(t)$ are shown in Figure 6.17(b).

Comments: A computer program, such as MatlabTM, can be used to visualize the result of a Fourier series approximation. Figure 6.17(c) shows one such approximation.

EXAMPLE 6.4 Computation of Fourier Series Coefficients

Problem

Compute the complete Fourier series expansion of the pulse waveform shown in Figure 6.18(a) for $\tau/T = 0.2$. Plot the spectrum of the signal.

Solution

Known Quantities: Amplitude and period of pulse train waveform.

Find: Fourier series coefficients a_n and b_n ; Fourier spectrum.

Schematics, Diagrams, Circuits, and Given Data: The function is periodic, with period $T = 1$ s, peak amplitude $A = 1$.

Assumptions: None.

Analysis: The function in Figure 6.18(a) is neither odd nor even. Thus, all the a_n and b_n coefficients are needed. First, express $x(t)$ over one period:

$$x(t) = \begin{cases} A & 0 \leq t < \tau \\ 0 & \tau \leq t < T \end{cases}$$

Then evaluate the integrals of equations 6.34 through 6.36:

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T A dt = \frac{A}{5} \\
 a_n &= \frac{2}{T} \int_0^T A \cos\left(n \frac{2\pi}{T} t\right) dt = \frac{2}{T} \int_0^{\tau} A \cos\left(n \frac{2\pi}{T} t\right) dt + \int_{\tau}^T 0 \cos\left(n \frac{2\pi}{T} t\right) dt \\
 &= \frac{2}{T} \frac{AT}{2n\pi} \sin\left(\frac{2n\pi}{T} t\right) \Big|_0^{\tau} \\
 &= \frac{2}{T} \frac{AT}{2n\pi} \left[\sin\left(\frac{2n\pi}{5}\right) - 0 \right] = \frac{A}{n\pi} \sin\left(\frac{2n\pi}{5}\right) \\
 b_n &= \frac{2}{T} \int_0^T A \sin\left(n \frac{2\pi}{T} t\right) dt = \frac{2}{T} \frac{AT}{2n\pi} \left[-\cos\left(\frac{2n\pi}{T} t\right) \right]_0^{\tau} \\
 &= \frac{2}{T} \frac{AT}{2n\pi} \left[-\cos\left(\frac{2n\pi}{5}\right) + \cos(0) \right] = \frac{A}{n\pi} \left[1 - \cos\left(\frac{2n\pi}{5}\right) \right]
 \end{aligned}$$

To compute the spectrum of the signal, apply equation 6.30:

$$\begin{aligned}
 c_n &= \sqrt{a_n^2 + b_n^2} = \sqrt{\left[\frac{A}{n\pi} \sin\left(\frac{2n\pi}{5}\right) \right]^2 + \left\{ \frac{A}{n\pi} \left[1 - \cos\left(\frac{2n\pi}{5}\right) \right] \right\}^2} \\
 \theta_n &= \cot^{-1} \left(\frac{b_n}{a_n} \right) = \cot^{-1} \left\{ \frac{(A/n\pi)[1 - \cos(2n\pi/5)]}{(A/n\pi) \sin(2n\pi/5)} \right\}
 \end{aligned}$$

The frequency spectrum of $x(t)$ (magnitude and phase) is shown in Figure 6.18(b).

Table 6.1 lists the first seven coefficients in both forms.

Comments: A Fourier series approximation with the first 10 frequency components included can be generated and visualized with MatlabTM. Figure 6.18(c) shows the result.

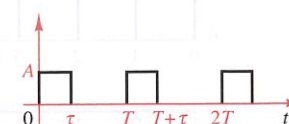


Figure 6.18 (a) Pulse train

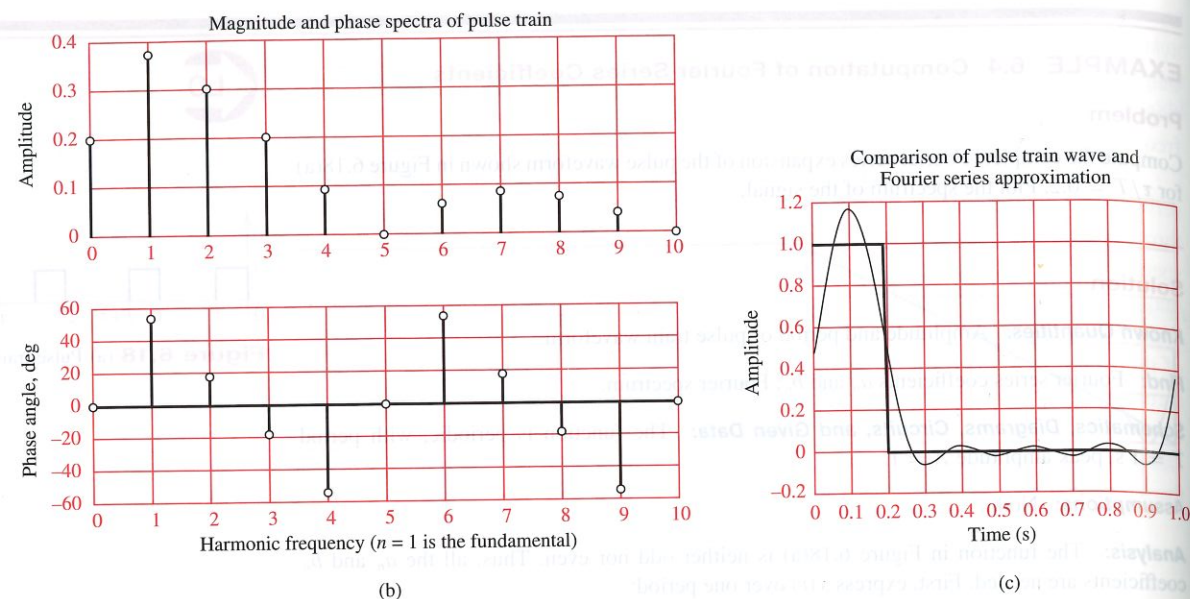


Figure 6.18 (b) signal spectrum; (c) approximation obtained using 11 Fourier coefficients

Table 6.1 Fourier coefficients of pulse train

n	a_n	b_n	c_n	θ_n (deg)
0	0.2	0	0.2	
1	0.3027	0.2199	0.3742	54
2	0.0935	0.2879	0.3027	18
3	-0.0624	0.1919	0.2018	-18
4	-0.0757	0.0550	0.0935	-54
5	0	0	0	0
6	0.0505	0.0367	0.0624	54
7	0.0267	0.0823	0.0865	18
8	-0.0234	0.0720	0.0757	-18
9	-0.0336	0.0244	0.0416	-54
10	0	0	0	0



EXAMPLE 6.5 Computation of Fourier Series Coefficients

Problem

Compute all the coefficients of the Fourier series expansion for the signal $x(t) = 1.5 \cos(100t)$.

Solution

Known Quantities: Expression of signal waveform.

Find: Fourier series coefficients a_n and b_n .

Schematics, Diagrams, Circuits, and Given Data: The function is periodic, with period $T = 2\pi/100$ and peak amplitude of 1.5.

Assumptions: None.

Analysis: This function is already in Fourier series form since it contains only sinusoidal terms! We recognize the following parameters: $\omega_0 = 100$; $a_0 = 0$; $a_1 = 1.5$; $b_1 = 0$; etc. (all other a_n and b_n coefficients are zero). Expressing the coefficients in magnitude-phase form, we have

$$c_1 = 1.5 \quad \text{and} \quad \theta_1 = \frac{\pi}{2}$$

EXAMPLE 6.6 Response of Linear System to a Periodic Input

Problem

A linear system with $\mathbf{H}(j\omega) = 2/(0.2j\omega + 1)$ is excited by the sawtooth waveform of Example 6.3 with $T = 0.25$ s and $A = 2$.

Solution

Known Quantities: $T = 0.25$ s; $A = 2$.

Find: Output $y(t)$ in response to input $x(t)$.

Assumptions: The waveform is well approximated by the first two terms of its Fourier series.

Analysis: The Fourier approximation of the sawtooth waveform of Example 6.3 is:

$$x(t) = \frac{2A}{\pi} \sin\left(\frac{2\pi}{0.25}t\right) + \frac{A}{\pi} \sin\left(\frac{4\pi}{0.25}t\right) = \frac{4}{\pi} \sin(8\pi t) + \frac{2}{\pi} \sin(16\pi t)$$

Then:

$$c_1 = \sqrt{a_1^2 + b_1^2} = |b_1| = \frac{4}{\pi} \quad \omega_1 = 1\omega_0 = 8\pi$$

and

$$c_2 = \sqrt{a_2^2 + b_2^2} = |b_2| = \frac{2}{\pi} \quad \omega_2 = 2\omega_0 = 16\pi$$

The frequency response of the system can then be expressed in magnitude and phase form:

$$\mathbf{H}(j\omega) = \frac{2}{0.2j\omega + 1} = |\mathbf{H}(j\omega)| \angle \mathbf{H}(j\omega) = \frac{2}{\sqrt{(0.2\omega)^2 + 1}} \angle \left(-\arctan \frac{\omega}{5}\right)$$

The magnitude and phase of the frequency response are shown in Figure 6.19(a). Observe that the system is excited only at the frequencies $\omega_1 = 8\pi = 25.1$ rad/s and $\omega_2 = 16\pi = 50.2$ rad/s. The frequency response of the system can be evaluated at these frequencies either graphically, as in Figure 6.19(a), or analytically, as shown here.

$$|\mathbf{H}(j\omega_1)| = \frac{2}{\sqrt{(0.2\omega_1)^2 + 1}} = 0.3902 \angle \mathbf{H}(j\omega_1) = -1.37 \text{ rad} = -78.75^\circ$$

$$|\mathbf{H}(j\omega_2)| = \frac{2}{\sqrt{(0.2\omega_2)^2 + 1}} = 0.1980 \angle \mathbf{H}(j\omega_2) = -1.47 \text{ rad} = -84.32^\circ$$

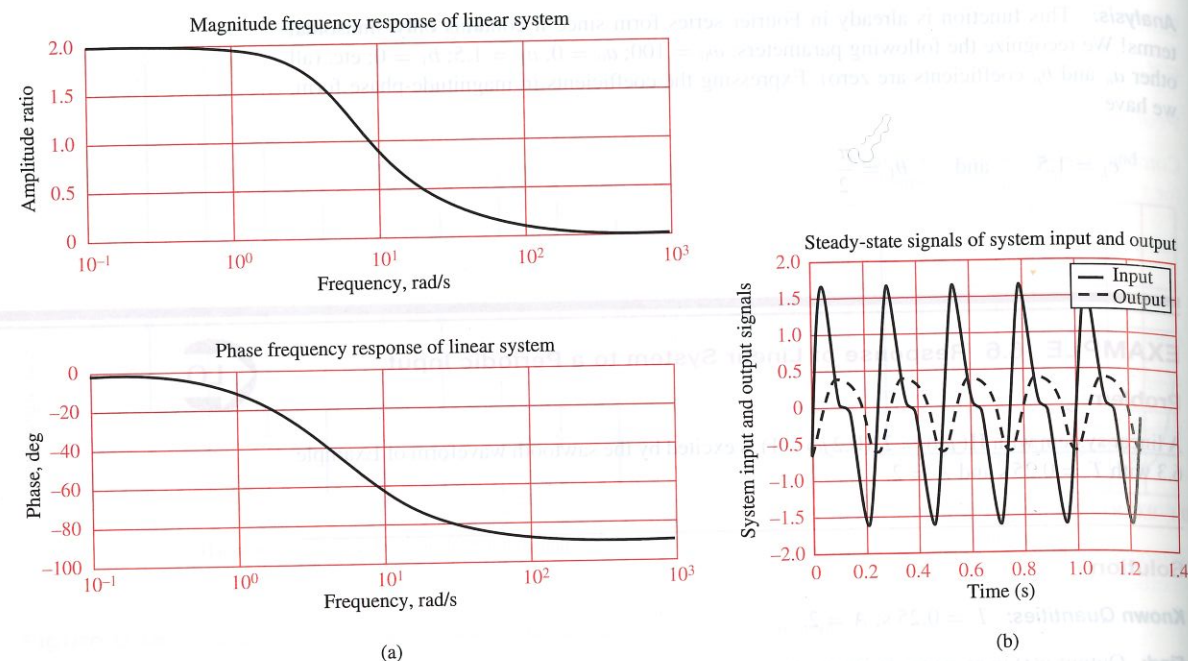


Figure 6.19 (a) Frequency response of linear system; (b) input and output waveforms

Finally, compute the steady-state periodic output of the system:

$$y(t) = \sum_{n=1}^2 |\mathbf{H}(j\omega_n)| c_n \sin[\omega_n t + \theta_n + \angle \mathbf{H}(j\omega_n)]$$

$$= 0.3902 \times \frac{4}{\pi} \sin(8\pi t - 1.37) + 0.1980 \times \frac{2}{\pi} \sin(16\pi t - 1.47)$$

The input and output signals for the system are plotted in Figure 6.19(b). Note how the first two components of the Fourier series of the sawtooth waveform of Example 6.3 provide a coarse approximation of the waveform. Given the frequency response of the system used in this example, would the accuracy of the computed response increase if higher-frequency components ($n > 2$) were included in the Fourier series approximation?

Comments: Matlab™ has built-in functions to perform calculations of linear systems. It can also be used to approximate any input as a Fourier series.

CHECK YOUR UNDERSTANDING

How would the spectrum plot of Figure 6.17(b) change if the period of the waveform changed from 1 to 0.1 s?

Answer: The fundamental frequency and all harmonics would increase tenfold; the shape is unchanged.

CHECK YOUR UNDERSTANDING

If the duty cycle of the pulse train in Figure 6.18(a) is changed to $\tau/T = 0.25$, which of the Fourier coefficients are now zero?

Answer: $n = 4, 8$

CHECK YOUR UNDERSTANDING

Determine the a_n and b_n Fourier coefficients of the signal $y(t) = 1.5 \cos(100t + \pi/4)$. (Hint: Use trigonometric identities to expand the cosine function.)

Answer: $a_0 = 0, a_1 = 1.0607, b_1 = 1.0607$. All other coefficients are zero.

CHECK YOUR UNDERSTANDING

Extend the result of Example 6.6 to include three frequency components. What are the amplitude and phase of the components of $y(t)$ at the frequency $3\omega_0$?

Answer: magnitude = 0.0562, phase = -1.505 rad = -86.2°

6.3 LOW- AND HIGH-PASS FILTERS

There are many practical applications that involve filters of one kind or another. Modern sunglasses filter out eye-damaging ultraviolet radiation and reduce the intensity of sunlight reaching the eyes. The suspension system of an automobile filters out road noise and reduces the impact of potholes on passengers. An analogous concept applies to electric circuits: it is possible to *attenuate* (i.e., reduce in amplitude) or altogether eliminate signals of unwanted frequencies, such as those that may be caused by electromagnetic interference (EMI).

Low-Pass Filters

Figure 6.20 depicts a simple **RC filter** and denotes its input and output voltages, respectively, by V_i and V_o . The frequency response for the filter may be obtained by considering the function

$$H(j\omega) = \frac{V_o}{V_i}(j\omega) \quad (6.39)$$

RC low-pass filter. The circuit preserves lower frequencies while attenuating the frequencies above the cutoff frequency $\omega_0 = 1/RC$. The voltages V_i and V_o are the filter input and output voltages, respectively.

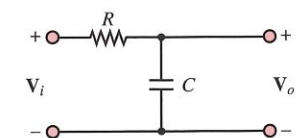


Figure 6.20 A simple RC filter

and noting that the output voltage may be expressed as a function of the input voltage by means of a voltage divider, as follows:

$$\mathbf{V}_o(j\omega) = \mathbf{V}_i(j\omega) \frac{1/j\omega C}{R + 1/j\omega C} = \mathbf{V}_i(j\omega) \frac{1}{1 + j\omega RC} \quad (6.40)$$

Thus, the frequency response of the RC filter is

$$\frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)} = \frac{1}{1 + j\omega RC} \quad (6.41)$$

An immediate observation upon studying this frequency response is that if the signal frequency ω is zero, the value of the frequency response function is 1. That is, the filter is passing all the input. Why? To answer this question, we note that at $\omega = 0$, the impedance of the capacitor, $1/j\omega C$, becomes infinite. Thus, the capacitor acts as an open-circuit, and the output voltage equals the input:

$$\mathbf{V}_o(j\omega = 0) = \mathbf{V}_i(j\omega = 0) \quad (6.42)$$

Since a signal at sinusoidal frequency equal to zero is a DC signal, this filter circuit does not in any way affect DC voltages and currents. As the signal frequency increases, the magnitude of the frequency response decreases since the denominator increases with ω . More precisely, equations 6.43 to 6.46 describe the magnitude and phase of the frequency response of the RC filter:

$$\begin{aligned} \mathbf{H}(j\omega) &= \frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)} = \frac{1}{1 + j\omega RC} \\ &= \frac{1}{\sqrt{1 + (\omega RC)^2}} e^{j \arctan(\omega RC/1)} \\ &= \frac{1}{\sqrt{1 + (\omega RC)^2}} \cdot e^{-j \arctan(\omega RC)} \end{aligned} \quad (6.43)$$

or

$$\mathbf{H}(j\omega) = |\mathbf{H}(j\omega)| e^{j\angle \mathbf{H}(j\omega)} \quad (6.44)$$

with

$$|\mathbf{H}(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}} \quad (6.45)$$

and

$$\angle \mathbf{H}(j\omega) = -\arctan(\omega RC) = -\arctan \frac{\omega}{\omega_0} \quad (6.46)$$

with

$$\omega_0 = \frac{1}{RC} \quad (6.47)$$

The simplest way to envision the effect of the filter is to think of the phasor voltage $\mathbf{V}_i = V_i e^{j\phi_i}$ scaled by a factor of $|\mathbf{H}|$ and shifted by a phase angle $\angle \mathbf{H}$ by the filter at each frequency, so that the resultant output is given by the phasor $\mathbf{V}_o e^{j\phi_o}$, with

$$\begin{aligned} \mathbf{V}_o &= |\mathbf{H}| \cdot \mathbf{V}_i \\ \phi_o &= \angle \mathbf{H} + \phi_i \end{aligned} \quad (6.48)$$

and where $|\mathbf{H}|$ and $\angle \mathbf{H}$ are functions of frequency. The frequency ω_0 is called the **cutoff frequency** of the filter and, as will presently be shown, gives an indication of the filtering characteristics of the circuit.

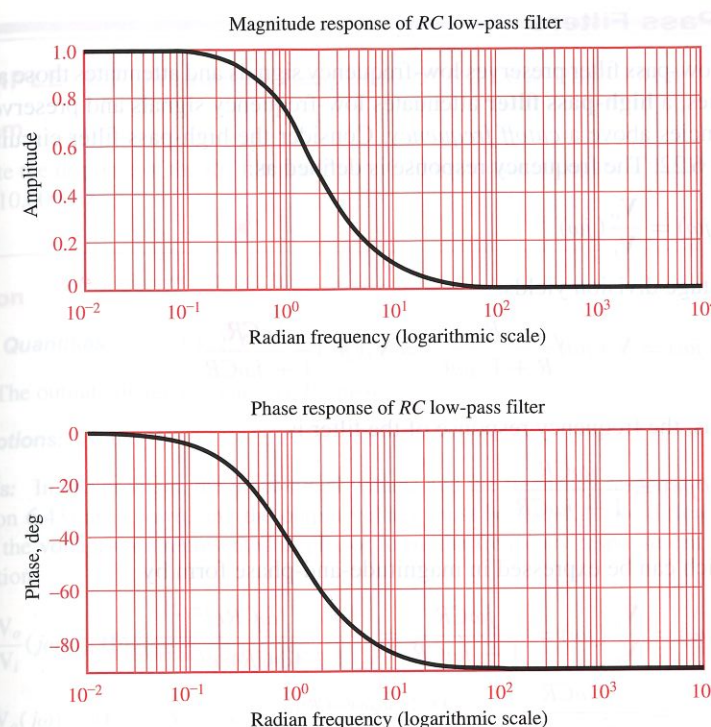


Figure 6.21 Magnitude and phase response plots for RC filter

It is customary to represent $\mathbf{H}(j\omega)$ in two separate plots, representing $|\mathbf{H}|$ and $\angle \mathbf{H}$ as functions of ω . These are shown in Figure 6.21 in normalized form, that is, with $|\mathbf{H}|$ and $\angle \mathbf{H}$ plotted versus ω/ω_0 , corresponding to a cutoff frequency $\omega_0 = 1$ rad/s. Note that, in the plot, the frequency axis has been scaled logarithmically. This is a common practice in electrical engineering because it enables viewing a very broad range of frequencies on the same plot without excessively compressing the low-frequency end of the plot. The frequency response plots of Figure 6.21 are commonly employed to describe the frequency response of a circuit since they can provide a clear idea at a glance of the effect of a filter on an excitation signal. For example, the RC filter of Figure 6.16 has the property of “passing” signals at low frequencies ($\omega \ll 1/RC$) and of filtering out signals at high frequencies ($\omega \gg 1/RC$). This type of filter is called a **low-pass filter**. The cutoff frequency $\omega = 1/RC$ has a special significance in that it represents—approximately—the point where the filter begins to filter out the higher-frequency signals. The value of $|\mathbf{H}(j\omega)|$ at the cutoff frequency is $1/\sqrt{2} = 0.707$. Note how the cutoff frequency depends exclusively on the values of R and C . Therefore, one can adjust the filter response as desired simply by selecting appropriate values for C and R , and therefore one can choose the desired filtering characteristics.

Practical low-pass filters are often much more complex than simple RC combinations. The synthesis of such advanced filter networks is beyond the scope of this book; however, the implementation of some commonly used filters is discussed in Chapters 8 and 15, in connection with the operational amplifier.

RC high-pass filter. The circuit preserves higher frequencies while attenuating the frequencies below the cutoff frequency $\omega_0 = 1/RC$.

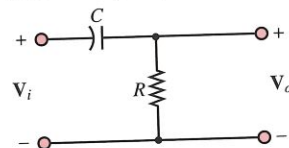


Figure 6.22 High-pass filter

High-Pass Filters

Just as a low-pass filter preserves low-frequency signals and attenuates those at higher frequencies, a **high-pass filter** attenuates low-frequency signals and preserves those at frequencies above a *cutoff frequency*. Consider the high-pass filter circuit shown in Figure 6.22. The frequency response is defined as:

$$\mathbf{H}(j\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i}(j\omega)$$

Voltage division yields:

$$\mathbf{V}_o(j\omega) = \mathbf{V}_i(j\omega) \frac{R}{R + 1/j\omega C} = \mathbf{V}_i(j\omega) \frac{j\omega CR}{1 + j\omega CR} \quad (6.49)$$

Thus, the frequency response of the filter is

$$\frac{\mathbf{V}_o}{\mathbf{V}_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR} \quad (6.50)$$

which can be expressed in magnitude-and-phase form by

$$\begin{aligned} \mathbf{H}(j\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i}(j\omega) &= \frac{j\omega CR}{1 + j\omega CR} = \frac{\omega CR e^{j\pi/2}}{\sqrt{1 + (\omega CR)^2} e^{j \arctan(\omega CR/1)}} \\ &= \frac{\omega CR}{\sqrt{1 + (\omega CR)^2}} \cdot e^{j[\pi/2 - \arctan(\omega CR)]} \end{aligned} \quad (6.51)$$

or

$$\mathbf{H}(j\omega) = |\mathbf{H}|e^{j\angle\mathbf{H}}$$

with

$$\begin{aligned} |\mathbf{H}(j\omega)| &= \frac{\omega CR}{\sqrt{1 + (\omega CR)^2}} \quad (6.52) \\ \angle\mathbf{H}(j\omega) &= 90^\circ - \arctan(\omega CR) \end{aligned}$$

You can verify by inspection that the amplitude response of the high-pass filter will be zero at $\omega = 0$ and will asymptotically approach 1 as ω approaches infinity while the phase shift is $\pi/2$ at $\omega = 0$ and tends to zero for increasing ω . Amplitude-and-phase response curves for the high-pass filter are shown in Figure 6.23. These plots have been normalized to have the filter cutoff frequency $\omega_0 = 1$ rad/s. Note that, once again, it is possible to define a cutoff frequency at $\omega_0 = 1/RC$ in the same way as was done for the low-pass filter.

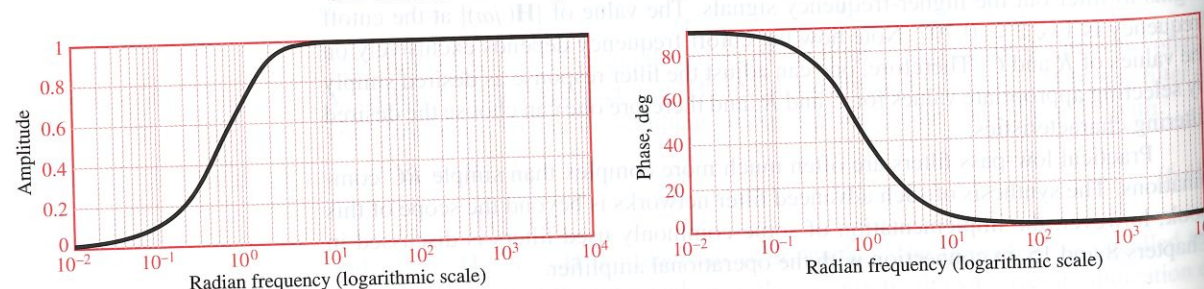


Figure 6.23 Frequency response of a high-pass filter

EXAMPLE 6.7 Frequency Response of RC Low-Pass Filter

Problem

Compute the response of the RC filter of Figure 6.20 to sinusoidal inputs at the frequencies of 60 and 10,000 Hz.

Solution

Known Quantities: $R = 1 \text{ k}\Omega$; $C = 0.47 \text{ }\mu\text{F}$; $v_i(t) = 5 \cos(\omega t) \text{ V}$.

Find: The output voltage $v_o(t)$ at each frequency.

Assumptions: None.

Analysis: In this problem, the input signal voltage and the frequency response of the circuit (equation 6.43) are known, and the output voltage must be found at two different frequencies. If the voltages are represented in phasor form, the frequency response can be used for calculation:

$$\frac{\mathbf{V}_o}{\mathbf{V}_i}(j\omega) = \mathbf{H}_V(j\omega) = \frac{1}{1 + j\omega CR}$$

$$\mathbf{V}_o(j\omega) = \mathbf{H}_V(j\omega)\mathbf{V}_i(j\omega) = \frac{1}{1 + j\omega CR} \mathbf{V}_i(j\omega)$$

The cutoff frequency of the filter is $\omega_0 = 1/RC = 2,128 \text{ rad/s}$ such that the expression for the frequency response in the form of equations 6.45 and 6.46 is:

$$\mathbf{H}_V(j\omega) = \frac{1}{1 + j\omega/\omega_0} \quad |\mathbf{H}_V(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}} \quad \angle\mathbf{H}(j\omega) = -\arctan\left(\frac{\omega}{\omega_0}\right)$$

Next, recognize that at $\omega = 60 \text{ Hz} = 120\pi \text{ rad/s}$, the ratio $\omega/\omega_0 = 0.177$. At $\omega = 10,000 \text{ Hz} = 20,000\pi$, $\omega/\omega_0 = 29.5$. Thus, the output voltage at each frequency can be computed as follows:

$$\mathbf{V}_o(\omega = 2\pi 60) = \frac{1}{1 + j0.177} \mathbf{V}_i(\omega = 2\pi 60) = (0.985 \times 5) \angle -0.175 \text{ V}$$

$$\mathbf{V}_o(\omega = 2\pi 10,000) = \frac{1}{1 + j29.5} \mathbf{V}_i(\omega = 2\pi 10,000) = (0.0339 \times 5) \angle -1.537 \text{ V}$$

And finally we write the time-domain response for each frequency:

$$v_o(t) = 4.923 \cos(2\pi 60t - 0.175) \text{ V} \quad \text{at } \omega = 2\pi 60 \text{ rad/s}$$

$$v_o(t) = 0.169 \cos(2\pi 10,000t - 1.537) \text{ V} \quad \text{at } \omega = 2\pi 10,000 \text{ rad/s}$$

The magnitude and phase responses of the filter are plotted in Figure 6.24. It should be evident from these plots that only the low-frequency components of the signal are passed by the filter. This low-pass filter would pass only the *bass range* of the audio spectrum.

Comments: Approximate answers can be obtained quickly from the magnitude and phase plots of Figure 6.18. Simply multiply the input voltage amplitude (5 V) by the amplitude response at each frequency, and read off the phase shift at each frequency. The results should be close to the ones computed above.

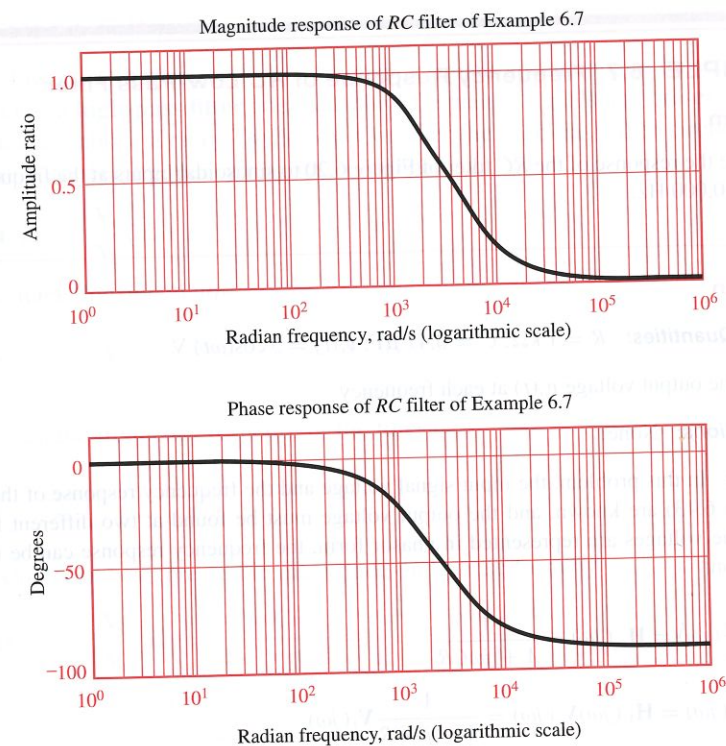


Figure 6.24 Response of RC filter of Example 6.7

**EXAMPLE 6.8 A Realistic RC Low-Pass Filter Application****Problem**

Determine the frequency response function V_o/V_s and its frequency response from the network shown in Figure 6.25.

Solution

Known Quantities: $R_s = 50 \Omega$; $R_1 = 200 \Omega$; $R_o = 500 \Omega$; $C = 10 \mu\text{F}$.

Find: The frequency response function V_o/V_s , its frequency response, and the output voltage $v_o(t)$ at given frequencies.

Assumptions: None.

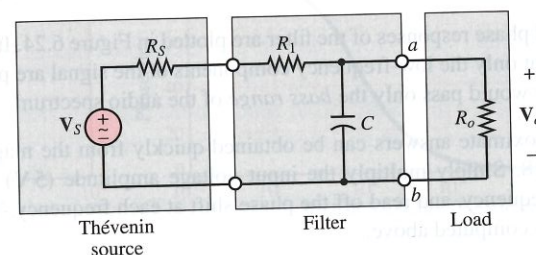


Figure 6.25 RC filter inserted in a circuit

Analysis: Figure 6.26 represents a more realistic filtering circuit, in that an RC low-pass filter is inserted between the source and load circuits. The Thévenin equivalent impedance seen by the load is:

$$Z_T = Z_C \parallel (R_1 + R_s) = \frac{1/j\omega C(R_1 + R_s)}{R_1 + R_s + 1/j\omega C}$$

Multiply the numerator and denominator by $j\omega C$ to obtain:

$$Z_T = \frac{R_1 + R_s}{1 + (j\omega)(R_1 + R_s)C}$$

Apply voltage division to find the Thévenin (open-circuit) voltage V_T across terminals a and b .

$$V_T = V_s \frac{1/j\omega C}{1/j\omega C + R_1 + R_s}$$

Again, multiply the numerator and denominator by $j\omega C$ to obtain:

$$V_T = V_s \frac{1}{1 + (j\omega)(R_1 + R_s)C}$$

Next, apply voltage division to find V_o .

$$V_o = V_T \frac{R_o}{R_o + Z_T}$$

Substitute for V_T and Z_T , and multiply the numerator and denominator by $[1 + (j\omega)(R_1 + R_s)C]$ to obtain:

$$V_o = V_s \frac{R_o}{R_o [1 + (j\omega)(R_1 + R_s)C] + (R_1 + R_s)}$$

Finally, divide both sides by V_s and factor $(R_o + R_1 + R_s)$ out of the denominator to find:

$$H_V(j\omega) = \frac{V_o}{V_s} = \frac{K}{1 + (j\omega)\tau}$$

where

$$K = \frac{R_o}{R_o + R_1 + R_s}$$

and

$$\tau = [R_o \parallel (R_1 + R_s)] C = \frac{R_o(R_1 + R_s)C}{R_o + R_1 + R_s}$$

Plug in values for the resistances and capacitance to find:

$$H(j\omega) = \frac{0.667}{1 + j(\omega/600)}$$

Comments: Notice that the time constant τ equals the capacitance times the Thévenin equivalent resistance seen by the capacitor. Thus, the effect of placing the RC low-pass filter in the midst of the circuit is to shift the filter's cutoff frequency from $1/R_1 C$ to $1/R_T C$. Also, note that the low-frequency amplitude of the frequency response function is simply K , which is $|V_o/V_s|$ when the capacitor is replaced with an open-circuit.

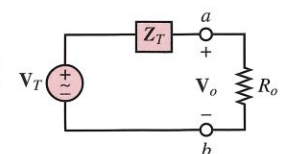


Figure 6.26 Equivalent circuit representation of Figure 6.25

EXAMPLE 6.9 Low-Pass Filter Attenuation**Problem**

The frequency response of a particular low-pass filter is described by the following frequency response function. At what frequency has the magnitude of the response fallen to 10 percent of its maximum?

$$H(j\omega) = \frac{K}{(j\omega/\omega_1 + 1)(j\omega/\omega_2 + 1)}$$



Solution

Known Quantities: Frequency response function of a filter.

Find: Frequency $\omega_{10\%}$ at which the response amplitude equals 10 percent of its maximum.

Schematics, Diagrams, Circuits, and Given Data: $\omega_1 = 100$; $\omega_2 = 1,000$.

Assumptions: None.

Analysis: The maximum amplitude of the frequency response function is K , which occurs as $\omega \rightarrow 0$. As frequency increases, the magnitude of the frequency response function decreases monotonically, which explains why the frequency response function describes a “low-pass” filter. At low frequencies, the input is “passed” to the output; however, at higher frequencies the output is a filtered (reduced) version of the input. To solve this problem, set the amplitude of the frequency response function equal to $0.1K$ and solve for ω , as follows:

$$|H(j\omega)| = \left| \frac{K}{(j\omega/\omega_1 + 1)(j\omega/\omega_2 + 1)} \right| = 0.1K$$

$$\frac{1}{\sqrt{(1 - \omega^2/\omega_1\omega_2)^2 + \omega^2(1/\omega_1 + 1/\omega_2)^2}} = 0.1$$

To simplify this expression introduce the dummy variable $\Omega = \omega^2$, and then invert and square both sides to obtain a quadratic equation in Ω :

$$\left(1 - \frac{\Omega}{\omega_1\omega_2}\right)^2 + \Omega\left(\frac{1}{\omega_1} + \frac{1}{\omega_2}\right)^2 = 100$$

$$\Omega^2 + \left[(\omega_1\omega_2)^2\left(\frac{1}{\omega_1} + \frac{1}{\omega_2}\right)^2 - 2\omega_1\omega_2\right]\Omega - 99(\omega_1\omega_2)^2 = 0$$

Plug in values for ω_1 and ω_2 and use the quadratic formula to solve for the two roots $\Omega = -1.6208 \times 10^6$ and $\Omega = 0.6108 \times 10^6$. Only the positive root has a physical meaning; thus, the solution is $\omega = \sqrt{\Omega} = 782$ rad/s. Figure 6.27(a) depicts the magnitude response of the filter. At a frequency roughly equal to 800 rad/s, the magnitude response is approximately 0.1. The phase response is shown in Figure 6.27(b).

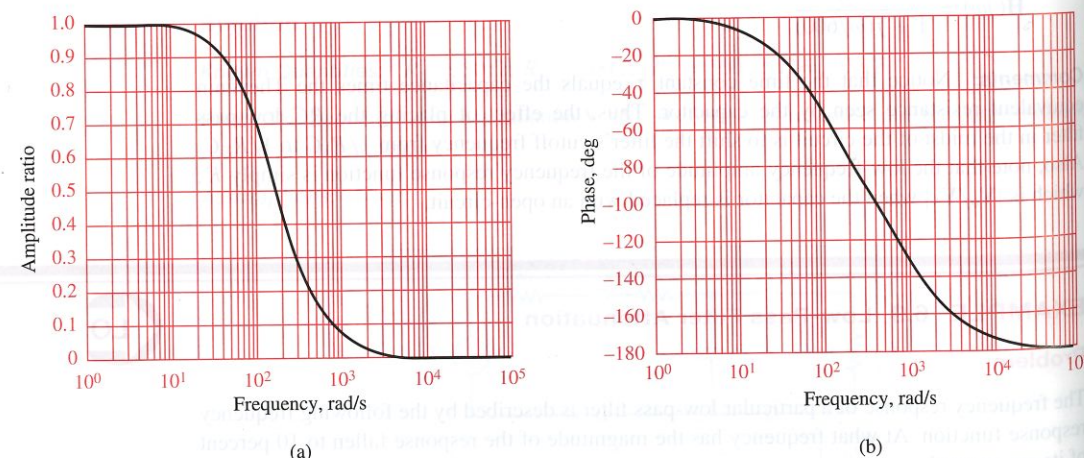


Figure 6.27 Frequency response of filter of Example 6.9. (a) Magnitude response; (b) phase response

EXAMPLE 6.10 Frequency Response of RC High-Pass Filter**Problem**

Compute the response of the RC high-pass filter depicted in Figure 6.28. Evaluate the response of the filter at $\omega_1 = 2\pi \times 100$ and $\omega_2 = 2\pi \times 10,000$ rad/s.

Solution

Known Quantities: $R = 200 \Omega$; $C = 0.199 \mu\text{F}$.

Find: The frequency response $H_V(j\omega)$.

Assumptions: None.

Analysis: The cutoff frequency of the high-pass filter is $\omega_0 = 1/RC = 25.126 \text{ kHz} = 2\pi \times 4,000$ rad/s, which is roughly halfway between ω_1 and ω_2 . The frequency response function for the circuit is given by equation 6.50:

$$H_V(j\omega) = \frac{V_o}{V_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR}$$

$$= \frac{\omega/\omega_0}{\sqrt{1 + (\omega/\omega_0)^2}} \angle \left[\frac{\pi}{2} - \arctan\left(\frac{\omega}{\omega_0}\right) \right]$$

The frequency response function can now be evaluated at ω_1 and ω_2 :

$$H_V(\omega = 2\pi \times 100) = \frac{100/4,000}{\sqrt{1 + (100/4,000)^2}} \angle \left[\frac{\pi}{2} - \arctan\left(\frac{100}{4,000}\right) \right] = 0.025 \angle 1.546$$

$$H_V(\omega = 2\pi \times 10,000) = \frac{10,000/4,000}{\sqrt{1 + (10,000/4,000)^2}} \angle \left[\frac{\pi}{2} - \arctan\left(\frac{10,000}{4,000}\right) \right]$$

$$= 0.929 \angle 0.38$$

These results indicate that the output is very small (2.5 percent) compared to the input at $\omega_1 \ll \omega_0$ while at $\omega_2 \gg \omega_0$ the output is comparable (92.9 percent) to the input. In general, the input is “passed” to the output at high frequencies ($\omega \gg \omega_0$) while at low frequencies ($\omega \ll \omega_0$) the output is a filtered (reduced) version of the input. The complete frequency response (amplitude and phase) is shown in Figure 6.29.

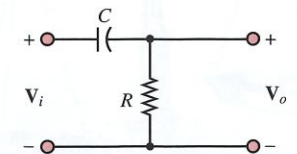


Figure 6.28 High-pass RC filter

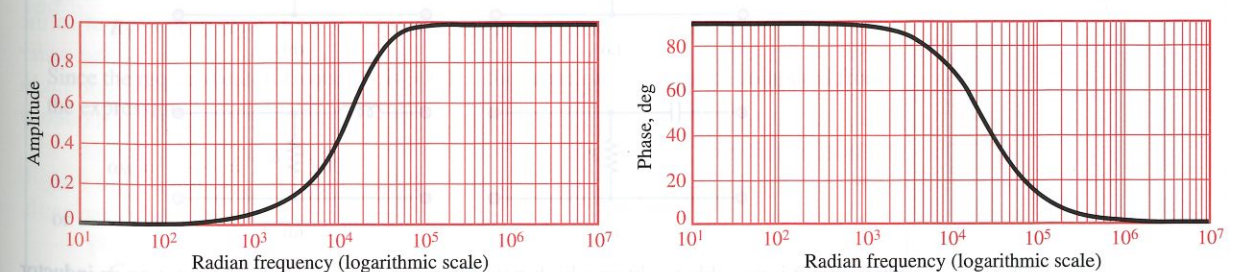


Figure 6.29 Response of high-pass RC filter of Example 6.10

Comments: With $\omega_0 = 2\pi \times 4,000$ (that is, 4,000 Hz), this filter would pass only the *treble* range of the audio frequency spectrum.

CHECK YOUR UNDERSTANDING

A simple RC low-pass filter is constructed using a $10\text{-}\mu\text{F}$ capacitor and a $2.2\text{-k}\Omega$ resistor. Over what range of frequencies will the output of the filter be within 1 percent of the input signal amplitude (i.e., when will $V_o \geq 0.99V_s$)?

Answer: $0 \leq \omega \leq 6.48 \text{ rad/s}$

CHECK YOUR UNDERSTANDING

In Figure 6.25, let $|V_s| = 1 \text{ V}$ with an internal resistance $R_s = 50 \Omega$. Assume $R = 1 \text{ k}\Omega$ and $C = 0.47$. Determine the cutoff frequency ω_0 for a load resistance $R_o = 470 \Omega$.

Answer: $\omega_0 = 6,553.3 \text{ rad/s}$

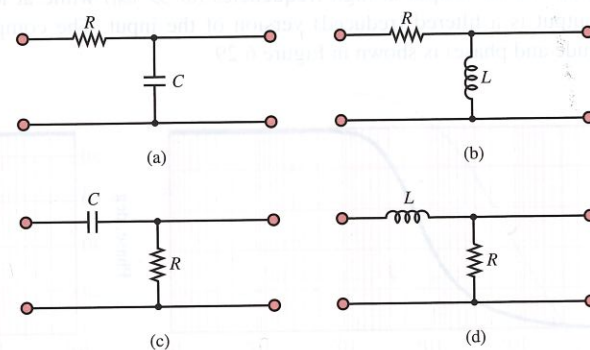
CHECK YOUR UNDERSTANDING

Use the phase response plot of Figure 6.27(b) to determine at which frequency the phase shift in the output signal (relative to the input signal) is equal to -90° .

Answer: $\omega = 300 \text{ rad/s}$ (approximately)

CHECK YOUR UNDERSTANDING

Determine the cutoff frequency for each of the four “prototype” filters shown below. Which are high-pass and which are low-pass?



Show that it is possible to obtain a high-pass filter response simply by substituting an inductor for the capacitor in the circuit of Figure 6.20. Derive the frequency response for the circuit.

Answers: (a) $\omega_0 = \frac{1}{RC}$ (low); (b) $\omega_0 = \frac{1}{L/R}$ (high); (c) $\omega_0 = \frac{1}{RC}$ (high); (d) $\omega_0 = \frac{1}{L/R}$ (low)

Wheatstone Bridge Filter

Problem:

The Wheatstone bridge circuit of Example 2.15 and Focus on Measurements box, “Wheatstone Bridge and Force Measurements” in Chapter 2 is used in a number of instrumentation applications, including the **measurement of force**. Figure 6.30 depicts the bridge circuit. When undesired noise and interference are present in a measurement, it is often appropriate to use a low-pass filter to reduce the effect of the noise. The capacitor that is connected to the output terminals of the bridge in Figure 6.30 constitutes an effective and simple low-pass filter, in conjunction with the bridge resistance. Assume that the average resistance of each leg of the bridge is 350Ω (a standard value for strain gauges) and that we desire to measure a sinusoidal force at a frequency of 30 Hz . From prior measurements, it has been determined that a filter with a cutoff frequency of 300 Hz is sufficient to reduce the effects of noise. Choose a capacitor that matches this filtering requirement.

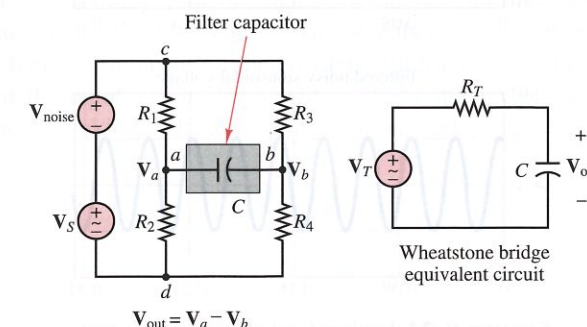


Figure 6.30 Wheatstone bridge with equivalent circuit and simple capacitive filter

Solution:

By evaluating the Thévenin equivalent circuit for the Wheatstone bridge, calculating the desired value for the filter capacitor becomes relatively simple, as illustrated on the right side of Figure 6.30. The Thévenin resistance for the bridge circuit may be computed by short-circuiting the two voltage sources and removing the capacitor placed across the load terminals:

$$R_T = R_1 \parallel R_2 + R_3 \parallel R_4 = 350 \parallel 350 + 350 \parallel 350 = 350 \Omega$$

Since the required cutoff frequency is 300 Hz , the capacitor value can be computed from the expression

$$\omega_0 = \frac{1}{R_T C} = 2\pi \times 300$$

or

$$C = \frac{1}{R_T \omega_0} = \frac{1}{350 \times 2\pi \times 300} = 1.51 \mu\text{F}$$

The frequency response of the bridge circuit is of the same form as equation 6.27:

$$\frac{V_{\text{out}}}{V_T}(j\omega) = \frac{1}{1 + j\omega C R_T}$$

(Continued)



FOCUS ON MEASUREMENTS

This response can be evaluated at the frequency of 30 Hz to verify that the attenuation and phase shift at the desired signal frequency are minimal:

$$\frac{V_{\text{out}}}{V_T}(j\omega = j2\pi \times 30) = \frac{1}{1 + j2\pi \times 30 \times 1.51 \times 10^{-6} \times 350} = 0.9951 \angle (-5.7^\circ)$$

Figure 6.31 depicts the appearance of a 30-Hz sinusoidal signal before and after the addition of the capacitor to the circuit.

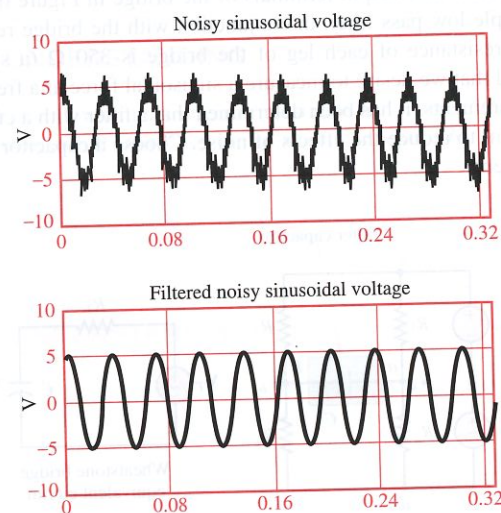


Figure 6.31 Unfiltered and filtered bridge output

6.4 BANDPASS FILTERS, RESONANCE, AND QUALITY FACTOR

Using the same principles and procedures as before, it is possible to derive a **bandpass filter** response for particular types of circuits. Such a filter passes the input to the output at frequencies *within* a certain range. The analysis of a simple *second-order* (i.e., two energy storage elements) bandpass filter is similar to that of low- and high-pass filters. Consider the circuit shown in Figure 6.32 and the designated frequency response function:

$$H(j\omega) = \frac{V_o}{V_i}(j\omega)$$

Apply voltage division to find:

$$\begin{aligned} V_o(j\omega) &= V_i(j\omega) \frac{R}{R + 1/j\omega C + j\omega L} \\ &= V_i(j\omega) \frac{j\omega CR}{1 + j\omega CR + (j\omega)^2 LC} \end{aligned} \quad (6.53)$$

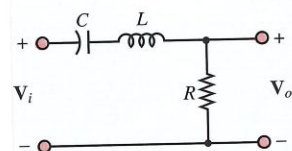


Figure 6.32 RLC bandpass filter.

Thus, the frequency response function is:

$$\frac{V_o}{V_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR + (j\omega)^2 LC} \quad (6.54)$$

Equation 6.50 can often be factored into the form

$$\frac{V_o}{V_i}(j\omega) = \frac{jA\omega}{(j\omega/\omega_1 + 1)(j\omega/\omega_2 + 1)} \quad (6.55)$$

where ω_1 and ω_2 are the two frequencies that determine the **passband** (or **bandwidth**) of the filter—that is, the frequency range over which the filter “passes” the input signal—and A is a constant that results from the factoring. An immediate observation we can make is that if the signal frequency ω is zero, the response of the filter is equal to zero since at $\omega = 0$ the impedance of the capacitor $1/j\omega C$ becomes infinite. Thus, the capacitor acts as an open-circuit, and the output voltage equals zero. Further, we note that the filter output in response to an input signal at sinusoidal frequency approaching infinity is again equal to zero. This result can be verified by considering that as ω approaches infinity, the impedance of the inductor becomes infinite, that is, an open-circuit. Thus, the filter cannot pass signals at very high frequencies. In an intermediate band of frequencies, the bandpass filter circuit will provide a variable attenuation of the input signal, dependent on the frequency of the excitation. This may be verified by taking a closer look at equation 6.53:

$$\begin{aligned} H(j\omega) &= \frac{V_o}{V_i}(j\omega) = \frac{jA\omega}{(j\omega/\omega_1 + 1)(j\omega/\omega_2 + 1)} \\ &= \frac{A\omega e^{j\pi/2}}{\sqrt{1 + (\omega/\omega_1)^2} \sqrt{1 + (\omega/\omega_2)^2} e^{j \arctan(\omega/\omega_1)} e^{j \arctan(\omega/\omega_2)}} \\ &= \frac{A\omega}{\sqrt{[1 + (\omega/\omega_1)^2][1 + (\omega/\omega_2)^2]}} e^{j[\pi/2 - \arctan(\omega/\omega_1) - \arctan(\omega/\omega_2)]} \end{aligned} \quad (6.56)$$

Equation 6.54 is of the form $H(j\omega) = |H|e^{j\angle H}$, with

$$|H(j\omega)| = \frac{A\omega}{\sqrt{[1 + (\omega/\omega_1)^2][1 + (\omega/\omega_2)^2]}} \quad (6.57)$$

and

$$\angle H(j\omega) = \frac{\pi}{2} - \arctan \frac{\omega}{\omega_1} - \arctan \frac{\omega}{\omega_2}$$

The magnitude and phase plots for the frequency response of the bandpass filter of Figure 6.32 are shown in Figure 6.33. These plots have been normalized to have the filter passband centered at the frequency $\omega = 1$ rad/s.

The frequency response plots of Figure 6.33 suggest that, in some sense, the bandpass filter acts as a combination of a high-pass and a low-pass filter. As illustrated in the previous cases, it should be evident that one can adjust the filter response as desired simply by selecting appropriate values for L , C , and R .

Resonance and Bandwidth

The response of second-order filters can be explained more generally by rewriting the frequency response function of the second-order bandpass filter of Figure 6.32 in

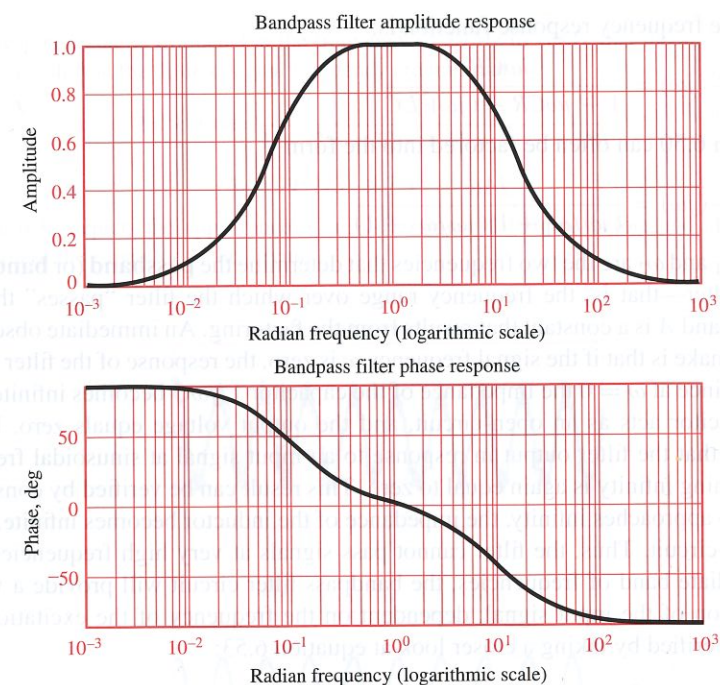


Figure 6.33 Frequency response of RLC bandpass filter

the following forms:

$$\begin{aligned}
 \frac{V_o(j\omega)}{V_i(j\omega)} &= \frac{j\omega CR}{LC(j\omega)^2 + j\omega CR + 1} \\
 &= \frac{(2\zeta/\omega_n)j\omega}{(j\omega/\omega_n)^2 + (2\zeta/\omega_n)j\omega + 1} \\
 &= \frac{(1/Q\omega_n)j\omega}{(j\omega/\omega_n)^2 + (1/Q\omega_n)j\omega + 1}
 \end{aligned} \quad (6.58)$$

with the following definitions:²



$$\begin{aligned}
 \omega_n &= \sqrt{\frac{1}{LC}} = \text{natural or resonant frequency} \\
 Q &= \frac{1}{2\zeta} = \frac{1}{\omega_n CR} = \omega_n \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \text{quality factor} \\
 \zeta &= \frac{1}{2Q} = \frac{R}{2} \sqrt{\frac{C}{L}} = \text{damping ratio}
 \end{aligned} \quad (6.59)$$

²If you have already studied the section on second-order transient response in Chapter 5, you will recognize the parameters ζ and ω_n .

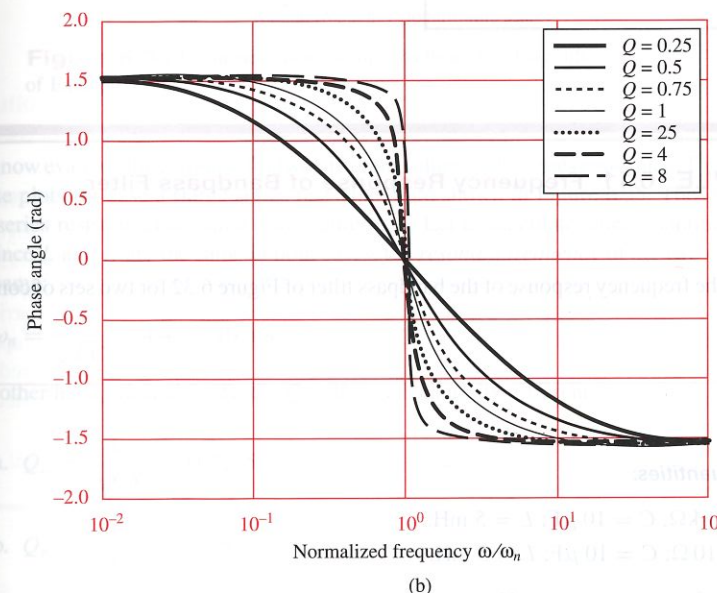
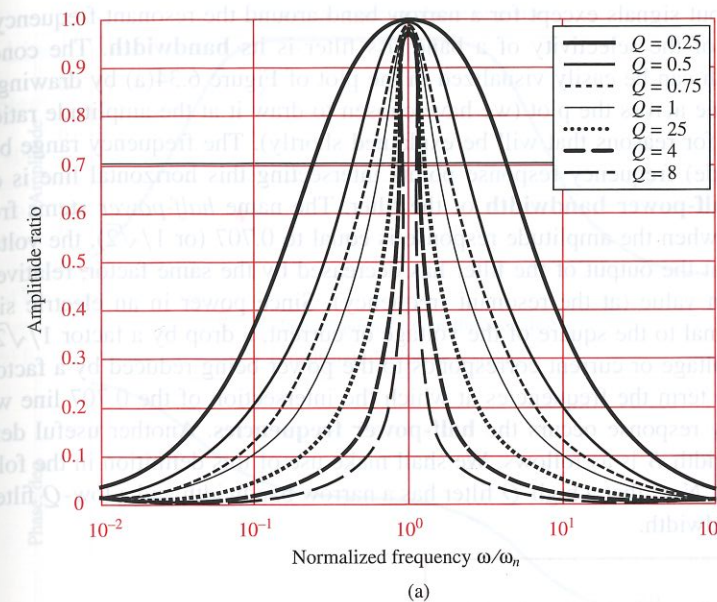


Figure 6.34 (a) Normalized magnitude response of second-order bandpass filter; (b) normalized phase response of second-order bandpass filter

Figure 6.34 depicts the normalized frequency response (magnitude and phase) of the second-order bandpass filter for $\omega_n = 1$ and various values of Q (and ζ). The peak displayed in the frequency response around the frequency ω_n is called a **resonant peak**, and ω_n is the **resonant frequency**. Note that as the **quality factor** Q increases, the sharpness of the resonance increases and the filter becomes increasingly **selective** (i.e., it has the ability to filter out most frequency components

of the input signals except for a narrow band around the resonant frequency). One measure of the selectivity of a bandpass filter is its **bandwidth**. The concept of bandwidth can be easily visualized in the plot of Figure 6.34(a) by drawing a horizontal line across the plot (we have chosen to draw it at the amplitude ratio value of 0.707 for reasons that will be explained shortly). The frequency range between (magnitude) frequency response points intersecting this horizontal line is defined as the **half-power bandwidth** of the filter. The name *half-power* stems from the fact that when the amplitude response is equal to 0.707 (or $1/\sqrt{2}$), the voltage (or current) at the output of the filter has decreased by the same factor, relative to the maximum value (at the resonant frequency). Since power in an electric signal is proportional to the square of the voltage or current, a drop by a factor $1/\sqrt{2}$ in the output voltage or current corresponds to the power being reduced by a factor of $\frac{1}{2}$. Thus, we term the frequencies at which the intersection of the 0.707 line with the frequency response occurs the **half-power frequencies**. Another useful definition of bandwidth B is as follows. We shall make use of this definition in the following examples. Note that a high- Q filter has a narrow bandwidth and a low- Q filter has a wide bandwidth.



$$B = \frac{\omega_n}{Q} \quad \text{Bandwidth}$$

(6.60)



EXAMPLE 6.11 Frequency Response of Bandpass Filter

Problem

Compute the frequency response of the bandpass filter of Figure 6.32 for two sets of component values.

Solution

Known Quantities:

- $R = 1 \text{ k}\Omega$; $C = 10 \text{ }\mu\text{F}$; $L = 5 \text{ mH}$.
- $R = 10 \text{ }\Omega$; $C = 10 \text{ }\mu\text{F}$; $L = 5 \text{ mH}$.

Find: The frequency response $\mathbf{H}_V(j\omega)$.

Assumptions: None.

Analysis: We write the frequency response of the bandpass filter as in equation 6.52:

$$\begin{aligned} \mathbf{H}_V(j\omega) &= \frac{\mathbf{V}_o}{\mathbf{V}_i}(j\omega) = \frac{j\omega CR}{1 + j\omega CR + (j\omega)^2 LC} \\ &= \frac{\omega CR}{\sqrt{(1 - \omega^2 LC)^2 + (\omega CR)^2}} \angle \left[\frac{\pi}{2} - \arctan \left(\frac{\omega CR}{1 - \omega^2 LC} \right) \right] \end{aligned}$$

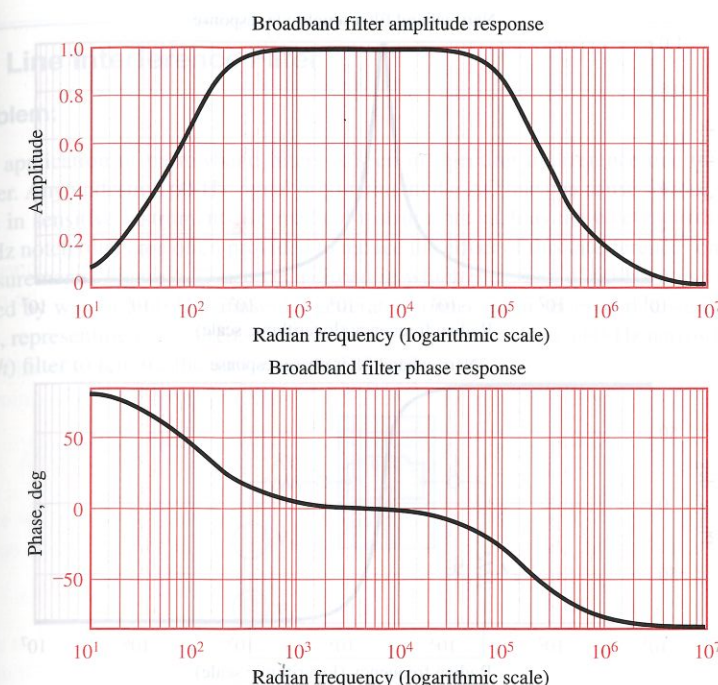


Figure 6.35 Frequency response of broadband bandpass filter of Example 6.11

We can now evaluate the response for two different values of the series resistance. The frequency response plots for case a (large series resistance) are shown in Figure 6.35. Those for case b (small series resistance) are shown in Figure 6.36. Let us calculate some quantities for each case. Since L and C are the same in both cases, the *resonant frequency* of the two circuits will be the same:

$$\omega_n = \frac{1}{\sqrt{LC}} = 4.47 \times 10^3 \text{ rad/s}$$

On the other hand, the *quality factor* Q will be substantially different:

$$\text{a. } Q_a = \frac{1}{\omega_n CR} \approx 0.02235$$

$$\text{b. } Q_b = \frac{1}{\omega_n CR} \approx 2.235$$

From these values of Q we can calculate the approximate bandwidth of the two filters:

$$B_a = \frac{\omega_n}{Q_a} \approx 200,000 \text{ rad/s} \quad \text{case a}$$

$$B_b = \frac{\omega_n}{Q_b} \approx 2,000 \text{ rad/s} \quad \text{case b}$$

The frequency response plots in Figures 6.35 and 6.36 confirm these observations.

Comments: It should be apparent that while at the higher and lower frequencies most of the amplitude of the input signal is filtered from the output, at the midband frequency (4,500 rad/s) most of the input signal amplitude passes through the filter. The first bandpass filter analyzed in this example would “pass” the *midband range* of the audio spectrum while the second would

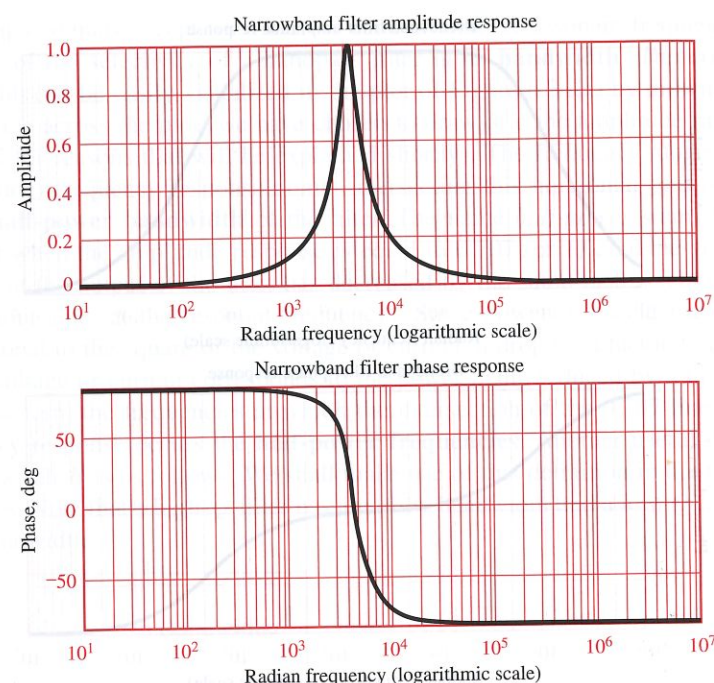


Figure 6.36 Frequency response of narrowband bandpass filter of Example 6.11

pass only a very narrow band of frequencies around the **center frequency** of 4,500 rad/s. Such narrowband filters find application in **tuning circuits**, such as those employed in conventional AM radios (although at frequencies much higher than that of the present example). In a tuning circuit, a narrowband filter is used to tune in a frequency associated with the **carrier wave** of a radio station (e.g., for a station found at a setting of AM 820, the carrier wave transmitted by the radio station is at a frequency of 820 kHz). By using a variable capacitor, it is possible to tune in a range of carrier frequencies and therefore select the preferred station. Other circuits are then used to decode the actual speech or music signal modulated on the carrier wave.

CHECK YOUR UNDERSTANDING

Compute the frequencies ω_1 and ω_2 for the bandpass filter of Example 6.11 (with $R = 1 \text{ k}\Omega$) by equating the magnitude of the bandpass filter frequency response to $1/\sqrt{2}$. The result is a quadratic equation in ω , which can be solved for two frequencies, known as the **half-power frequencies**.

Answer: $\omega_1 = 99.95 \text{ rad/s}$; $\omega_2 = 200.1 \text{ krad/s}$

AC Line Interference Filter

Problem:

One application of narrowband filters is seen in rejecting interference due to AC line power. Any undesired 60-Hz signal originating in the AC line can cause serious interference in sensitive instruments. In medical instruments such as the **electrocardiograph**, 60-Hz notch filters are often provided to reduce the effect of this interference³ on cardiac measurements. Figure 6.37 depicts a circuit in which the effect of 60-Hz noise is represented by way of a 60-Hz sinusoidal generator connected in series with a signal source (V_S), representing the desired signal. In this example we design a 60-Hz narrowband (or *notch*) filter to remove the unwanted 60-Hz noise.

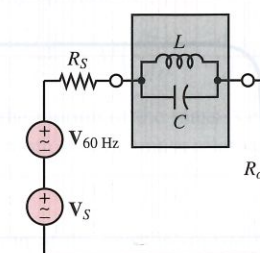


Figure 6.37 60-Hz notch filter

Solution:

Known Quantities — $R_S = 50 \Omega$.

Find — Appropriate values of L and C for the notch filter.

Assumptions — None.

Analysis — To determine the appropriate capacitor and inductor values, we write the expression for the notch filter impedance:

$$\begin{aligned} Z_{\parallel} &= Z_o \parallel Z_C = \frac{j\omega L / j\omega C}{j\omega L + 1/j\omega C} \\ &= \frac{j\omega L}{1 - \omega^2 LC} \end{aligned}$$

Note that when $\omega^2 LC = 1$, the impedance of the circuit is infinite! The frequency

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

is the resonant frequency of the LC circuit. If this resonant frequency were selected to be equal to 60 Hz, then the series circuit would show an infinite impedance to 60-Hz currents and would therefore block the interference signal, while passing most of the other frequency components. We thus select values of L and C that result in $\omega_0 = 2\pi \times 60$. Let $L = 100 \text{ mH}$. Then

$$C = \frac{1}{\omega_0^2 L} = 70.36 \mu\text{F}$$

(Continued)



FOCUS ON MEASUREMENTS

³See the Focus on Measurements box: "Electrocardiogram Amplifier" in Chapter 8 and section 15.2 for further information on electrocardiograms and line noise, respectively.

The frequency response of the complete circuit is given below:

$$\begin{aligned} H_V(j\omega) &= \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{R_o}{R_S + R_o + Z_{\parallel}} \\ &= \frac{R_o}{R_S + R_o + j\omega L / (1 - \omega^2 LC)} \end{aligned}$$

and is plotted in Figure 6.38.

Comments—It is instructive to calculate the response of the notch filter at frequencies in the immediate neighborhood of 60 Hz, to verify the attenuation effect of the notch filter.

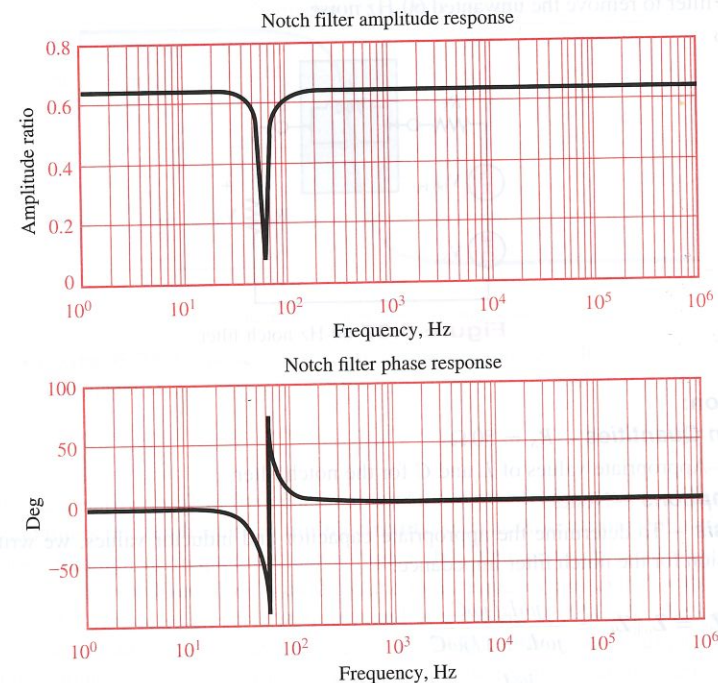
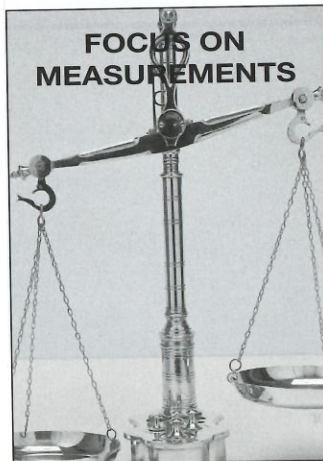


Figure 6.38 Frequency response of 60-Hz notch filter



Seismic Transducer

This example illustrates the application of the frequency response idea to a practical displacement transducer. The frequency response of a **seismic displacement transducer** is analyzed, and it is shown that there is an analogy between the equations describing the mechanical transducer and those that describe a second-order electric circuit.

The configuration of the transducer is shown in Figure 6.39. The transducer is housed in a case rigidly affixed to the surface of a body whose motion is to be measured. Thus, the case will experience the same displacement as the body, x_i . Inside the case, a small mass M rests on a spring characterized by stiffness K , placed in parallel with a damper B . The wiper arm of a potentiometer is connected to the floating mass M ; the potentiometer is attached to the transducer case, so that the voltage V_o is proportional to the relative displacement of the mass with respect to the case x_o .

(Continued)

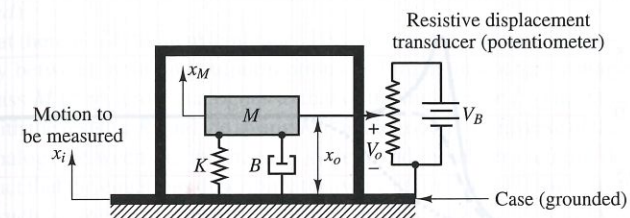


Figure 6.39 Seismic displacement transducer

The equation of motion for the mass-spring-damper system may be obtained by summing all the forces acting on mass M :

$$Kx_o + B \frac{dx_o}{dt} = M \frac{d^2x_M}{dt^2} = M \left(\frac{d^2x_i}{dt^2} - \frac{d^2x_o}{dt^2} \right)$$

where we have noted that the motion of the mass is equal to the difference between the motion of the case and the motion of the mass relative to the case itself; that is,

$$x_M = x_i - x_o$$

If we assume that the motion of the mass is sinusoidal, we may use phasor analysis to obtain the frequency response of the transducer by defining the phasor quantities

$$\mathbf{X}_i(j\omega) = |X_i|e^{j\phi_i} \quad \text{and} \quad \mathbf{X}_o(j\omega) = |X_o|e^{j\phi_o}$$

The assumption of a sinusoidal motion may be justified in light of the discussion of Fourier analysis in section 6.2. If we then recall (from Chapter 4) that taking the derivative of a phasor corresponds to multiplying the phasor by $j\omega$, we can rewrite the second-order differential equation as follows:

$$\begin{aligned} M(j\omega)^2 \mathbf{X}_o + B(j\omega) \mathbf{X}_o + K \mathbf{X}_o &= M(j\omega)^2 \mathbf{X}_i \\ (-\omega^2 M + j\omega B + K) \mathbf{X}_o &= -\omega^2 M \mathbf{X}_i \end{aligned}$$

and we can write an expression for the frequency response:

$$\frac{\mathbf{X}_o(j\omega)}{\mathbf{X}_i(j\omega)} = H(j\omega) = \frac{-\omega^2 M}{-\omega^2 M + j\omega B + K}$$

The frequency response of the transducer is plotted in Figure 6.40 for the component values $M = 0.005$ kg and $K = 1,000$ N/m and for three values of B :

- $B = 10$ N-s/m dotted line
- $B = 2$ N-s/m dashed line
- $B = 1$ N-s/m solid line

The transducer clearly displays a high-pass response, indicating that for a sufficiently high input signal frequency, the measured displacement (proportional to the voltage V_o) is equal to the input displacement x_i , which is the desired quantity. Note how sensitive the frequency response of the transducer is to changes in damping: as B changes from 2 to 1, a sharp **resonant peak** appears around the frequency $\omega = 316$ rad/s (approximately 50 Hz). As B increases to a value of 10, the amplitude response curve shifts to the right. Thus, this transducer, with the preferred damping given by $B = 2$, would be capable of correctly measuring displacements at frequencies above a minimum value, about

(Continued)

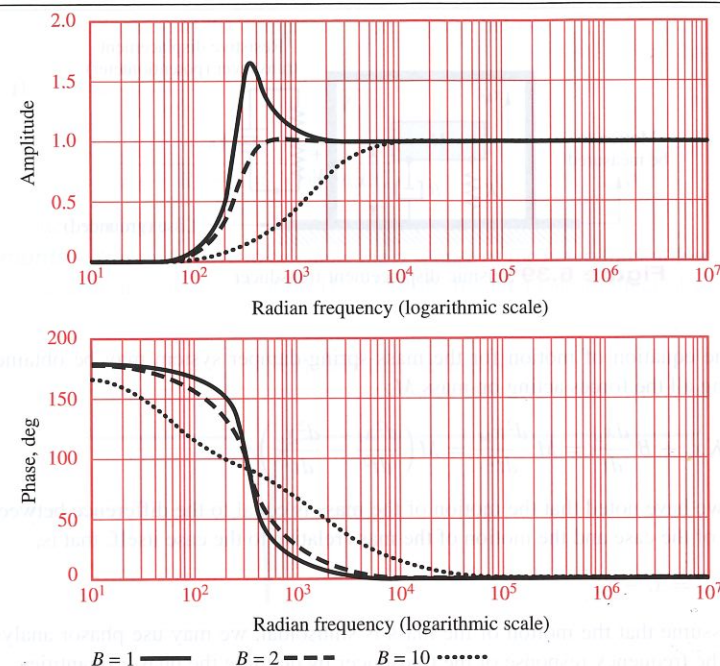


Figure 6.40 Frequency response of seismic transducer

1,000 rad/s (or 159 Hz). The choice of $B = 2$ as the preferred design may be explained by observing that, ideally, we would like to obtain a constant amplitude response at all frequencies. The magnitude response that most closely approximates the ideal case in Figure 6.40 corresponds to $B = 2$. This concept is commonly applied to a variety of **vibration measurements**.

We now illustrate how a second-order electric circuit will exhibit the same type of response as the seismic transducer. Consider the circuit shown in Figure 6.41. The frequency response for the circuit may be obtained by using the principles developed in the preceding sections:

$$\begin{aligned} \frac{V_o}{V_i}(j\omega) &= \frac{j\omega L}{R + 1/j\omega C + j\omega L} = \frac{(j\omega L)(j\omega C)}{j\omega CR + 1 + (j\omega L)(j\omega C)} \\ &= \frac{-\omega^2 L}{-\omega^2 L + j\omega R + 1/C} \end{aligned}$$

Comparing this expression with the frequency response of the seismic transducer,

$$\frac{X_o(j\omega)}{X_i(j\omega)} = H(j\omega) = \frac{-\omega^2 M}{-\omega^2 M + j\omega B + K}$$

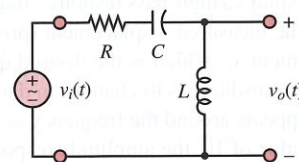


Figure 6.41 Electric circuit analog of the seismic transducer

(Continued)

(Concluded)

we find that there is a definite resemblance between the two. In fact, it is possible to draw an analogy between input and output motions and input and output voltages. Note also that the mass M plays a role analogous to that of the inductance L . The damper B acts in analogy with the resistor R ; and the spring K is analogous to the inverse of the capacitance, C . This analogy between the mechanical system and the electric circuit derives simply from the fact that the equations describing the two systems have the same form. Engineers often use such analogies to construct electrical *models*, or *analogs*, of physical systems. For example, to study the behavior of a large mechanical system, it might be easier and less costly to start by modeling the mechanical system with an inexpensive electric circuit and testing the model, rather than the full-scale mechanical system.

6.5 BODE PLOTS

Frequency response plots of linear systems are often displayed in the form of logarithmic plots, called **Bode plots** after the mathematician Hendrik W. Bode, where the horizontal axis represents frequency on a logarithmic scale (base 10) and the vertical axis represents either the amplitude or phase of the frequency response function. In Bode plots the amplitude is expressed in units of **decibels (dB)**, where

$$\left| \frac{A_o}{A_i} \right|_{\text{dB}} = 20 \log_{10} \left| \frac{A_o}{A_i} \right| = 20 \log_{10} \left| \frac{A_o}{A_i} \right| \quad (6.61)$$

While logarithmic plots may at first seem a daunting complication, they have two significant advantages:

1. The product of terms in a frequency response function becomes a sum of terms because $\log(ab/c) = \log(a) + \log(b) - \log(c)$. The advantage here is that Bode (logarithmic) plots can be constructed from the sum of individual plots of individual terms. Moreover, as was discussed in section 6.1, there are only four distinct types of terms present in any frequency response function:
 - a. A constant K .
 - b. Poles or zeros "at the origin" ($j\omega$).
 - c. Simple poles or zeros $(1 + j\omega\tau)$ or $(1 + j\omega/\omega_0)$.
 - d. Quadratic poles or zeros $[1 + j\omega\tau + (j\omega/\omega_n)^2]$.
2. The individual Bode plots of these four distinct terms are all well approximated by linear segments, which are readily summed to form the overall Bode plot of more complicated frequency response functions.

RC Low-Pass Filter Bode Plots

Consider, for example, the RC low-pass filter of Example 6.7 (Figure 6.20). The frequency response function is:

$$\frac{V_o}{V_i}(j\omega) = \frac{1}{1 + j\omega/\omega_0} = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}} \angle -\tan^{-1} \left(\frac{\omega}{\omega_0} \right) \quad (6.62)$$

where the circuit time constant is $\tau = RC = 1/\omega_0$ and ω_0 is the cutoff, or half-power, frequency of the filter. This frequency response function has a constant of value $K = 1$ and a simple pole with cutoff frequency $\omega_0 = 1/\tau = 1/RC$.

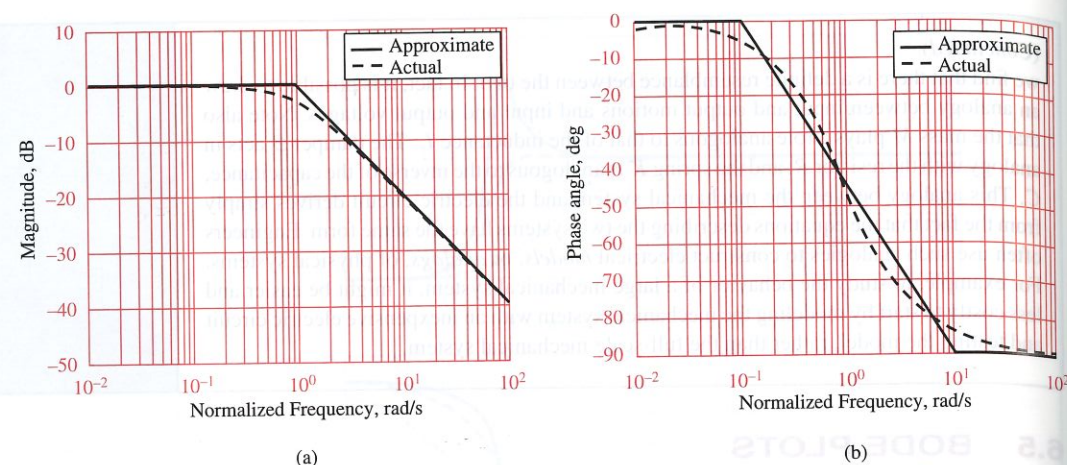


Figure 6.42 Bode plots for a low-pass RC filter; the frequency variable is normalized to ω/ω_0 . (a) Magnitude response; (b) phase angle response

Figure 6.42 shows the Bode magnitude and phase plots for the filter. The normalized frequency on the horizontal axis is $\omega\tau$. The magnitude plot is obtained from the logarithmic form of the absolute value of the frequency response function:

$$\left| \frac{V_o}{V_i} \right|_{dB} = 20 \log_{10} \frac{|K|}{|1 + j\omega\tau|} = 20 \log_{10} \frac{|K|}{|1 + j\omega/\omega_0|} \quad (6.63)$$

When $\omega \ll \omega_0$, the imaginary part of the simple pole is much smaller than its real part, such that $|1 + j\omega/\omega_0| \approx 1$. Then:

$$\left| \frac{V_o}{V_i} \right|_{dB} \approx 20 \log_{10} K - 20 \log_{10} 1 = 20 \log_{10} K \quad (dB) \quad (6.64)$$

Thus, at very low frequencies ($\omega \ll \omega_0$), equation 6.63 is well approximated by a straight line of zero slope, which is the *low-frequency asymptote* of the Bode magnitude plot.

When $\omega \gg \omega_0$, the imaginary part of the simple pole is much larger than its real part, such that $|1 + j\omega/\omega_0| \approx |j\omega/\omega_0| = (\omega/\omega_0)$. Then:

$$\begin{aligned} \left| \frac{V_o}{V_i} \right|_{dB} &\approx 20 \log_{10} K - 20 \log_{10} \frac{\omega}{\omega_0} \\ &\approx 20 \log_{10} K - 20 \log_{10} \omega + 20 \log_{10} \omega_0 \end{aligned} \quad (6.65)$$

Thus, at very high frequencies ($\omega \gg \omega_0$), equation 6.63 is well approximated by a straight line of -20 dB per decade slope that intercepts the $\log \omega$ axis at $\log \omega_0$. This line is the *high-frequency asymptote* of the Bode magnitude plot. A decade represents a factor of 10 change in frequency. Thus, a one decade increase in ω is equivalent to a unity change in $\log \omega$.

Finally, when $\omega = \omega_0$, the real and imaginary parts of the simple pole are equal, such that $|1 + j\omega/\omega_0| = |1 + j| = \sqrt{2}$. Then equation 6.63 becomes:

$$20 \log_{10} \frac{|K|}{|1 + j\omega/\omega_0|} = 20 \log_{10} K - 20 \log_{10} \sqrt{2} = 20 \log_{10} K - 3 \text{ dB} \quad (6.66)$$

Thus, the Bode magnitude plot of a first-order low-pass filter is approximated by two straight lines intersecting at ω_0 . Figure 6.42(a) clearly shows the approximation. The actual Bode magnitude plot is 3 dB lower than the approximate plot at $\omega = \omega_0$, the cutoff frequency.

The phase angle of the frequency response function $\angle(V_o/V_i) = -\tan^{-1}(\omega/\omega_0)$ has the following properties:

$$-\tan^{-1} \left(\frac{\omega}{\omega_0} \right) = \begin{cases} 0 & \text{when } \omega \rightarrow 0 \\ -\frac{\pi}{4} & \text{when } \omega = \omega_0 \\ -\frac{\pi}{2} & \text{when } \omega \rightarrow \infty \end{cases}$$

As a first approximation, the phase angle can be represented by three straight lines:

1. For $\omega < 0.1\omega_0$, $\angle(V_o/V_i) \approx 0$.
2. For $0.1\omega_0$ and $10\omega_0$, $\angle(V_o/V_i) \approx -(\pi/4) \log(10\omega/\omega_0)$.
3. For $\omega > 10\omega_0$, $\angle(V_o/V_i) \approx -\pi/2$.

These straight-line approximations are illustrated in Figure 6.42(b).

Table 6.2 lists the differences between the actual and approximate Bode magnitude and phase plots. Note that the maximum difference in magnitude is 3 dB at the cutoff frequency; thus, the cutoff is often called the **3-dB frequency** or the *half-power frequency*.

Table 6.2 Correction factors for asymptotic approximation of first-order filter

ω/ω_0	Magnitude response error, (dB)	Phase response error (deg)
0.1	0	-5.7
0.5	-1	4.9
1	-3	0
2	-1	-4.9
10	0	+5.7

RC High-Pass Filter Bode Plots

The case of an RC high-pass filter (see Figure 6.22) is analyzed in the same manner as was done for the RC low-pass filter. The frequency response function is:

$$\begin{aligned} \frac{V_o}{V_i} &= \frac{j\omega CR}{1 + j\omega CR} = \frac{j(\omega/\omega_0)}{1 + j(\omega/\omega_0)} \\ &= \frac{(\omega/\omega_0) \angle(\pi/2)}{\sqrt{1 + (\omega/\omega_0)^2} \angle \arctan(\omega/\omega_0)} \\ &= \frac{\omega/\omega_0}{\sqrt{1 + (\omega/\omega_0)^2}} \angle \left(\frac{\pi}{2} - \arctan \frac{\omega}{\omega_0} \right) \end{aligned} \quad (6.67)$$

Figure 6.43 depicts the Bode plots for equation 6.67, where the horizontal axis indicates the normalized frequency ω/ω_0 . Straight-line asymptotic approximations may again be determined easily at low and high frequencies. The results are very similar to those for the first-order low-pass filter. For $\omega < \omega_0$, the Bode magnitude

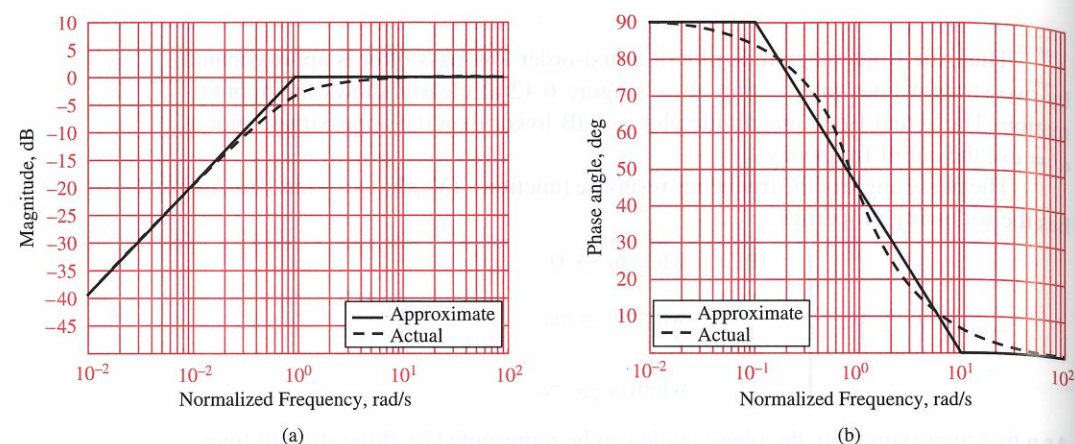


Figure 6.43 Bode plots for RC high-pass filter. (a) Magnitude response; (b) phase response

approximation intercepts the origin ($\omega = 1$) with a slope of +20 dB/decade. For $\omega > \omega_0$, the Bode magnitude approximation is 0 dB with zero slope. The straight-line approximations of the Bode phase plot are:

1. For $\omega < 0.1\omega_0$, $\angle(\mathbf{V}_o/\mathbf{V}_i) \approx \pi/2$.
2. For $0.1\omega_0$ and $10\omega_0$, $\angle(\mathbf{V}_o/\mathbf{V}_i) \approx -(\pi/4) \log(10\omega/\omega_0)$.
3. For $\omega > 10\omega_0$, $\angle(\mathbf{V}_o/\mathbf{V}_i) \approx 0$.

These straight-line approximations are illustrated in Figure 6.43(b).

Bode Plots of Higher-Order Filters

Bode plots of high-order systems may be obtained by combining Bode plots of factors of the higher-order frequency response function. Let, for example,

$$\mathbf{H}(j\omega) = \mathbf{H}_1(j\omega)\mathbf{H}_2(j\omega)\mathbf{H}_3(j\omega) \quad (6.68)$$

which can be expressed, in logarithmic form, as

$$|\mathbf{H}(j\omega)|_{\text{dB}} = |\mathbf{H}_1(j\omega)|_{\text{dB}} + |\mathbf{H}_2(j\omega)|_{\text{dB}} + |\mathbf{H}_3(j\omega)|_{\text{dB}} \quad (6.69)$$

and

$$\angle\mathbf{H}(j\omega) = \angle\mathbf{H}_1(j\omega) + \angle\mathbf{H}_2(j\omega) + \angle\mathbf{H}_3(j\omega) \quad (6.70)$$

Consider as an example the frequency response function

$$\mathbf{H}(j\omega) = \frac{j\omega + 5}{(j\omega + 10)(j\omega + 100)} \quad (6.71)$$

The first step in computing the asymptotic approximation consists of factoring each term in the expression so that it appears in the form $a_i(j\omega/\omega_i + 1)$, where the frequency ω_i corresponds to the appropriate 3-dB frequency, ω_1 , ω_2 , or ω_3 . For example, the function of equation 6.71 is rewritten as:

$$\begin{aligned} \mathbf{H}(j\omega) &= \frac{5(j\omega/5 + 1)}{10(j\omega/10 + 1)100(j\omega/100 + 1)} \\ &= \frac{0.005(j\omega/5 + 1)}{(j\omega/10 + 1)(j\omega/100 + 1)} = \frac{K(j\omega/\omega_1 + 1)}{(j\omega/\omega_2 + 1)(j\omega/\omega_3 + 1)} \end{aligned} \quad (6.72)$$

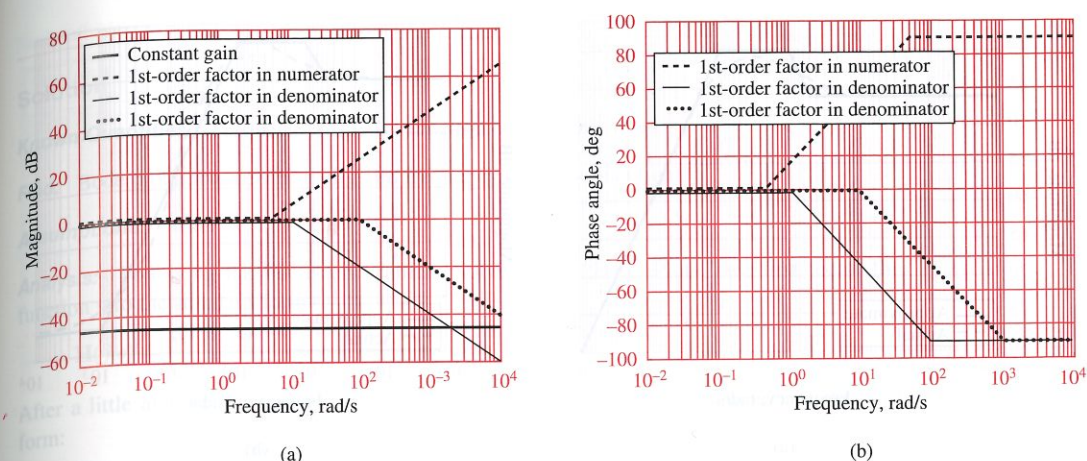


Figure 6.44 Bode plot approximation for a second-order frequency response function. (a) Straight-line approximation of magnitude response; (b) straight-line approximation of phase angle response

Equation 6.72 can now be expressed in logarithmic form:

$$\begin{aligned} |\mathbf{H}(j\omega)|_{\text{dB}} &= |0.005|_{\text{dB}} + \left| \frac{j\omega}{5} + 1 \right|_{\text{dB}} - \left| \frac{j\omega}{10} + 1 \right|_{\text{dB}} - \left| \frac{j\omega}{100} + 1 \right|_{\text{dB}} \\ \angle\mathbf{H}(j\omega) &= \angle 0.005 + \angle \left(\frac{j\omega}{5} + 1 \right) - \angle \left(\frac{j\omega}{10} + 1 \right) - \angle \left(\frac{j\omega}{100} + 1 \right) \end{aligned} \quad (6.73)$$

Each of the terms in the logarithmic magnitude expression can be plotted individually. The constant corresponds to the value -46 dB, plotted in Figure 6.44(a) as a line of zero slope. The numerator term, with a 3-dB frequency $\omega_1 = 5$, is expressed in the form of the first-order Bode plot of Figure 6.42(a), except for the fact that the slope of the line leaving the zero axis at $\omega_1 = 5$ is +20 dB/decade; each of the two denominator factors is similarly plotted as lines of slope -20 dB/decade, departing the zero axis at $\omega_2 = 10$ and $\omega_3 = 100$. You see that the individual factors are very easy to plot by inspection once the frequency response function has been normalized in the form of equation 6.69.

If we now consider the phase response portion of equation 6.73, we recognize that the first term, the phase angle of the constant, is always zero. The numerator first-order term, on the other hand, can be approximated as shown in Figure 6.42(b), that is, by drawing a straight line starting at $0.1\omega_1 = 0.5$, with slope $+\pi/4$ rad/decade (positive because this is a numerator factor) and ending at $10\omega_1 = 50$, where the asymptote $+\pi/2$ is reached. The two denominator terms have similar behavior, except for the fact that the slope is $-\pi/4$ and that the straight line with slope $-\pi/4$ rad/decade extends between the frequencies $0.1\omega_2$ and $10\omega_2$, and $0.1\omega_3$ and $10\omega_3$, respectively.

Figure 6.44 depicts the asymptotic approximations of the individual factors in equation 6.73, with the magnitude factors shown in Figure 6.44(a) and the phase factors in Figure 6.44(b). When all the asymptotic approximations are combined, the complete frequency response approximation is obtained. Figure 6.45 depicts the results of the asymptotic Bode approximation when compared with the actual frequency response functions.

You can see that once a frequency response function is factored into the appropriate form, it is relatively easy to sketch a good approximation of the Bode plot, even for higher-order frequency response functions. Examples 6.12 and 6.13 illustrate some additional details. The methodology is summarized in the box below.

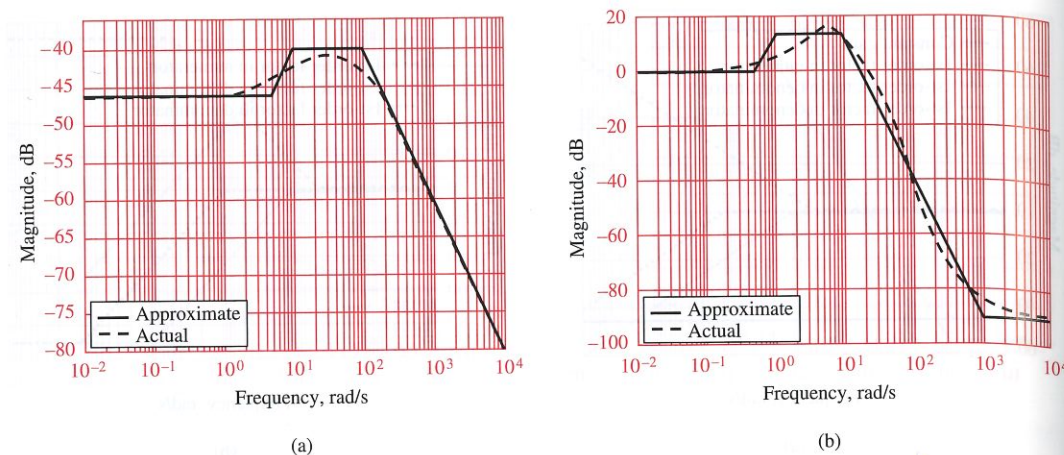


Figure 6.45 Comparison of Bode plot approximation with the actual frequency response function. (a) Magnitude response of second-order frequency response function; (b) phase angle response of second-order frequency response function.



FOCUS ON PROBLEM SOLVING

BODE PLOTS

This box illustrates the Bode plot asymptotic approximation construction procedure. The method assumes that there are no complex conjugate factors in the response and that both the numerator and denominator can be factored into first-order terms with real roots.

1. Express the frequency response function in factored form, resulting in an expression similar to equation 6.57:

$$H(j\omega) = \frac{K(j\omega/\omega_1 + 1) \cdots (j\omega/\omega_m + 1)}{(j\omega/\omega_{m+1} + 1) \cdots (j\omega/\omega_n + 1)}$$

2. Select the appropriate frequency range for the semilogarithmic plot, extending at least a decade below the lowest 3-dB frequency and a decade above the highest 3-dB frequency.
3. Sketch the magnitude and phase response asymptotic approximations for each of the first-order factors, using the techniques illustrated in Figures 6.42 to 6.45.
4. Add, graphically, the individual terms to obtain a composite response.
5. If desired, apply the correction factors of Table 6.2.

EXAMPLE 6.12 Bode Plot Approximation



Problem

Sketch the asymptotic approximation of the Bode plot for the frequency response function

$$H(j\omega) = \frac{0.1j\omega + 20}{2 \times 10^{-5}(j\omega)^3 + 0.1002(j\omega)^2 + j\omega}$$

Solution

Known Quantities: Frequency response function of a circuit.

Find: Bode plot approximation of given frequency response function.

Assumptions: None

Analysis: Following the Focus on Problem Solving box “Bode Plots,” we first factor the function into the standard form

$$H(j\omega) = \frac{K(j\omega/\omega_1 + 1) \cdots (j\omega/\omega_m + 1)}{(j\omega/\omega_{m+1} + 1) \cdots (j\omega/\omega_n + 1)}$$

After a little algebra, we can obtain the following frequency response function in standard form:

$$H(j\omega) = \frac{20(j\omega/200 + 1)}{j\omega(j\omega/10 + 1)(j\omega/5,000 + 1)}$$

We immediately notice that there is a factor of $j\omega$ in the denominator; this term needs to be treated somewhat differently. The Bode plot of the function $1/j\omega$ can be expressed in logarithmic form as follows:

$$\left| \frac{1}{j\omega} \right|_{dB} = -20 \log_{10} \frac{\omega}{1}$$

$$\angle \frac{1}{j\omega} = 0 - \frac{\pi}{2} = -\frac{\pi}{2}$$

That is, the magnitude of the denominator factor $j\omega$ is represented by a line with slope of -20 dB/decade intersecting the frequency (horizontal) axis at $\omega = 1$. Its phase response is a constant equal to $-\pi/2$.

Now we can sketch the magnitude and phase response of each of the individual first-order factors, as shown in Figure 6.46(a) and (b). The composite asymptotic approximations of the magnitude and phase responses are shown in Figure 6.47(a) and (b).

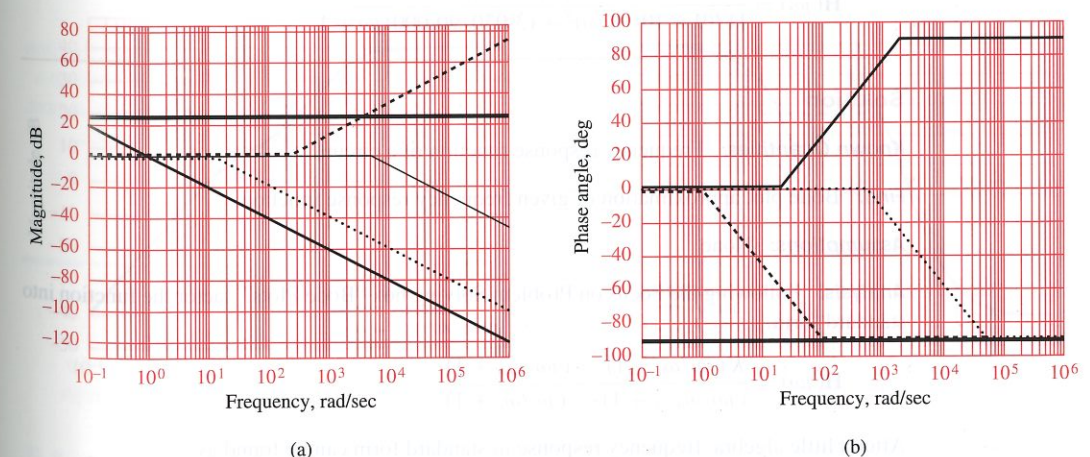


Figure 6.46 Approximate (asymptotic) frequency response of individual first-order terms. (a) Straight-line approximation of magnitude response; (b) straight-line approximation of phase angle response.

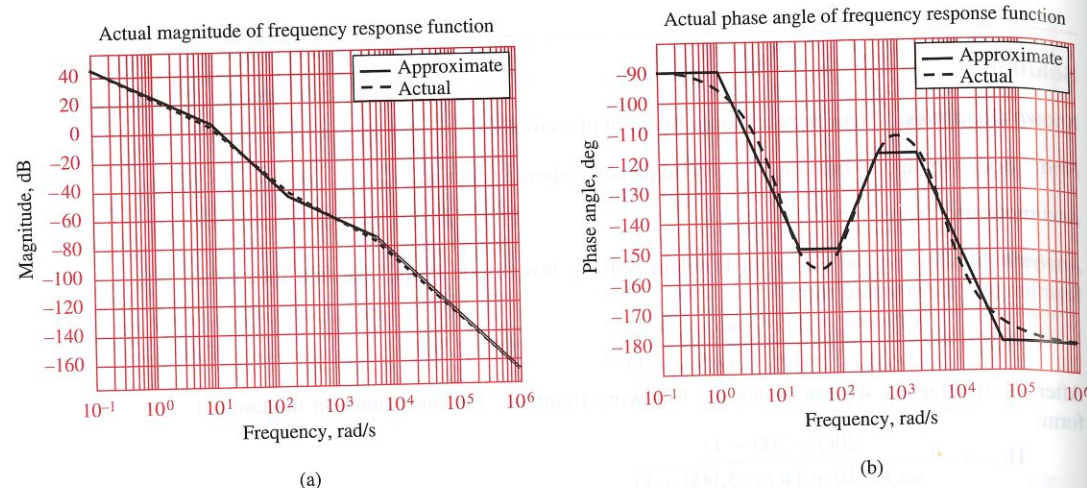


Figure 6.47 Comparison of approximate and exact frequency response. (a) Actual magnitude of frequency response function; (b) actual phase angle of frequency response function.

Comments: A computer program can be used to generate the Bode plot approximation shown in Figures 6.46 and 6.47. Note that the only real effort in generating the asymptotic approximation lies in the factoring of the frequency response function.



EXAMPLE 6.13 Bode Plot Approximation

Problem

Sketch the asymptotic approximation of the Bode plot for the frequency response function

$$H(j\omega) = \frac{10^{-3}(j\omega)^2 + 0.1j\omega}{[1/(9 \times 10^4)](j\omega)^2 + (3,030/90,000)j\omega + 1}$$

Solution

Known Quantities: Frequency response function of a circuit.

Find: Bode plot approximation of given frequency response function.

Assumptions: None

Analysis: Following the Focus on Problem Solving box “Bode Plots,” factor the function into standard form.

$$H(j\omega) = \frac{K(j\omega/\omega_1 + 1) \cdots (j\omega/\omega_m + 1)}{(j\omega/\omega_{m+1} + 1) \cdots (j\omega/\omega_n + 1)}$$

After a little algebra, frequency response in standard form can be found as:

$$H(j\omega) = \frac{0.1j\omega(j\omega/100 + 1)}{(j\omega/30 + 1)(j\omega/3,000 + 1)}$$

Notice the $j\omega$ factor in the numerator. This factor can be expressed in logarithmic form as:

$$|j\omega|_{dB} = 20 \log_{10} \frac{\omega}{1}$$

$$\angle j\omega = \frac{\pi}{2}$$

That is, the magnitude of $j\omega$ is represented by a line with slope +20 dB/decade intersecting the frequency (horizontal) axis at $\omega = 1$. The phase of the factor $j\omega$ is a constant and equal to $\pi/2$.

The magnitude and phase of each individual first-order factor can be sketched as shown in Figure 6.48(a) and (b). The composite asymptotic approximations of the magnitude and phase responses are shown in Figure 6.49(a) and (b).

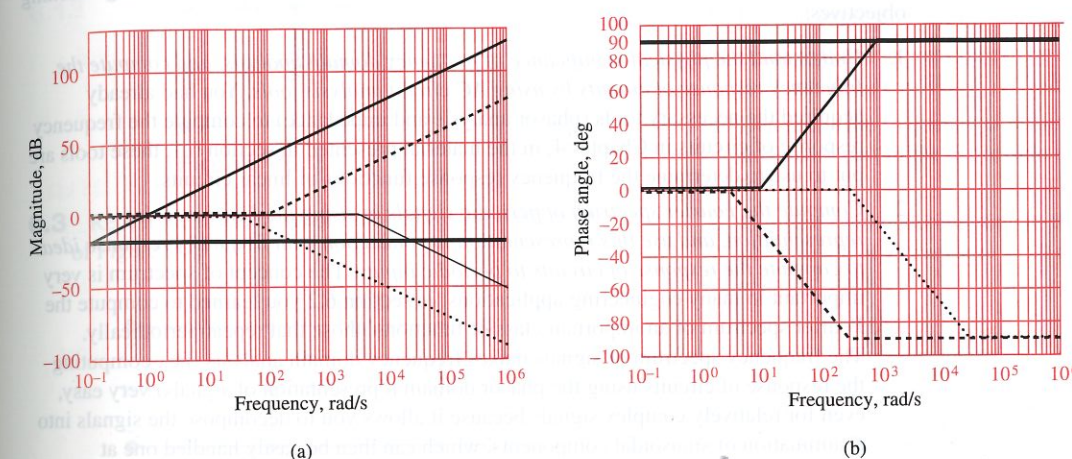


Figure 6.48 Approximate (asymptotic) frequency response of individual first-order terms. (a) Straight-line approximation of magnitude response; (b) straight-line approximation of phase angle response.

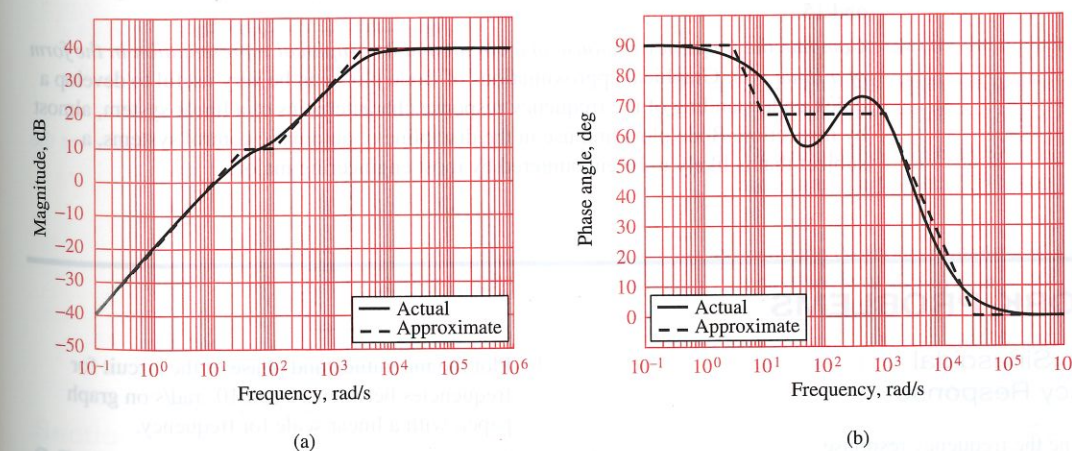


Figure 6.49 Comparison of approximate and exact frequency responses. (a) Actual magnitude of frequency response function; (b) actual phase angle of frequency response function.

Comments: Bode plots can be generated using Matlab™. Circuit simulation programs, such as B2Spice, can also generate Bode plots.

Conclusion

Chapter 6 focuses on the frequency response of linear circuits, and it is a natural extension of the material covered in Chapter 4. The concepts of the spectrum of a signal, obtained through the Fourier series representation for periodic signals, and of the frequency response of a filter are very useful ideas that extend well beyond electrical engineering. For example, civil, mechanical, and aeronautical engineering students who study the vibrations of structures and machinery will find that the same methods are employed in those fields.

Upon completing this chapter, you should have mastered the following learning objectives:

1. Understand the physical significance of frequency domain analysis, and compute the frequency response of circuits by using AC circuit analysis tools. You had already acquired the necessary tools (phasor analysis and impedance) to compute the frequency response of circuits in Chapter 4; in the material presented in section 6.1, these tools are put to use to determine the frequency response functions of linear circuits.
2. Compute the Fourier spectrum of periodic signals by using the Fourier series representation, and use this representation in connection with frequency response ideas to compute the response of circuits to periodic inputs. The concept of spectrum is very important in many engineering applications; in section 6.2 you learned to compute the Fourier spectrum of an important class of functions: those that repeat periodically. The frequency spectrum of signals makes frequency domain analysis (i.e., computing the response of circuits using the phasor domain representation of signals) very easy, even for relatively complex signals because it allows you to decompose the signals into a summation of sinusoidal components, which can then be easily handled one at a time.
3. Analyze simple first- and second-order electrical filters, and determine their frequency response and filtering properties. With the concept of frequency response firmly in hand, now you can analyze the behavior of electrical filters and study the frequency response characteristics of the most common types, that is, low-pass, high-pass, and bandpass filters. Filters are very useful devices and are explored in greater depth in Chapters 8 and 15.
4. Compute the frequency response of a circuit and its graphical representation in the form of a Bode plot. Graphical approximations of Bode plots can be very useful to develop a quick understanding of the frequency response characteristics of a linear system, almost by inspection. Bode plots find use in the discipline of automatic control systems, a subject that is likely to be encountered by most engineering majors.

HOMEWORK PROBLEMS

Section 6.1: Sinusoidal Frequency Response

- 6.1** a. Determine the frequency response $V_{out}(j\omega)/V_{in}(j\omega)$ for the circuit of Figure P6.1. Assume $L = 0.5$ H and $R = 200$ k Ω .

- b. Plot the magnitude and phase of the circuit for frequencies between 10 and 10⁷ rad/s on graph paper, with a linear scale for frequency.
- c. Repeat part b, using semilog paper. (Place the frequency on the logarithmic axis.)

- d. Plot the magnitude response on semilog paper with magnitude in decibels.

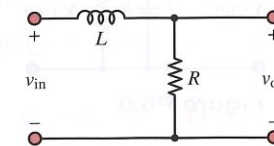


Figure P6.1

- 6.2** Repeat the instructions of Problem 6.1 for the circuit of Figure P6.2.

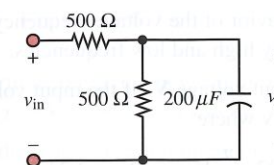


Figure P6.2

- 6.3** Repeat the instructions of Problem 6.1 for the circuit of Figure P6.3.

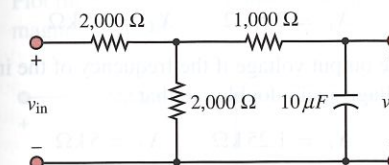


Figure P6.3

- 6.4** Repeat Problem 6.1 for the circuit of Figure P6.4. $R_1 = 300$ Ω , $R_2 = R_3 = 500$ Ω , $L = 4$ H, $C_1 = 40$ μ F, $C_2 = 160$ μ F.

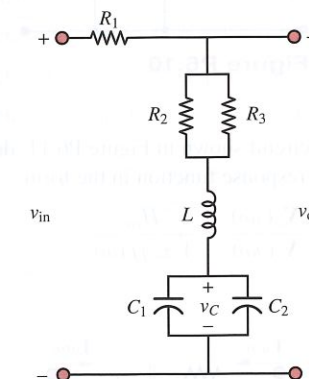


Figure P6.4

- 6.5** Determine the frequency response of the circuit of Figure P6.5, and generate frequency response plots. $R_1 = 20$ k Ω , $R_2 = 100$ k Ω , $L = 1$ H, $C = 100$ μ F.

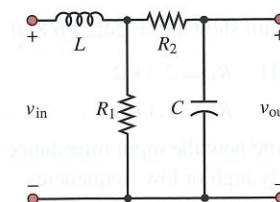


Figure P6.5

- 6.6** In the circuit shown in Figure P6.6, where $C = 0.5$ μ F and $R = 2$ k Ω ,
- a. Determine how the input impedance $Z(j\omega) = V_i(j\omega)/I_i(j\omega)$ behaves at extremely high and low frequencies.
 - b. Find an expression for the impedance.
 - c. Show that this expression can be manipulated into the form $Z(j\omega) = R[1 + j(1/\omega RC)]$.
 - d. Determine the frequency $\omega = \omega_C$ for which the imaginary part of the expression in part c is equal to 1.
 - e. Estimate (without computing it) the magnitude and phase angle of $Z(j\omega)$ at $\omega = 10$ rad/s and $\omega = 10^5$ rad/s.

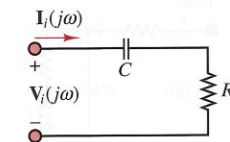


Figure P6.6

- 6.7** In the circuit shown in Figure P6.7, where $L = 2$ mH and $R = 2$ k Ω ,
- a. Determine how the input impedance $Z(j\omega) = V_i(j\omega)/I_i(j\omega)$ behaves at extremely high and low frequencies.
 - b. Find an expression for the impedance.
 - c. Show that this expression can be manipulated into the form $Z(j\omega) = R[1 + j(\omega L/R)]$.
 - d. Determine the frequency $\omega = \omega_C$ for which the imaginary part of the expression in part c is equal to 1.
 - e. Estimate (without computing it) the magnitude and phase angle of $Z(j\omega)$ at $\omega = 10^5$ rad/s, 10^6 rad/s, and 10^7 rad/s.

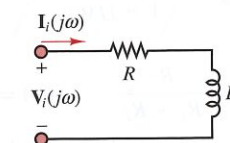


Figure P6.7

6.8 In the circuit shown in Figure P6.8, if

$$L = 190 \text{ mH} \quad R_1 = 2.3 \text{ k}\Omega$$

$$C = 55 \text{ nF} \quad R_2 = 1.1 \text{ k}\Omega$$

- Determine how the input impedance behaves at extremely high or low frequencies.
- Find an expression for the input impedance in the form

$$Z(j\omega) = Z_o \left[\frac{1 + jf_1(\omega)}{1 + jf_2(\omega)} \right]$$

$$Z_o = R_1 + \frac{L}{R_2 C}$$

$$f_1(\omega) = \frac{\omega^2 R_1 L C - R_1 - R_2}{\omega(R_1 R_2 C + L)}$$

$$f_2(\omega) = \frac{\omega^2 L C - 1}{\omega C R_2}$$

- Determine the four frequencies at which $f_1(\omega) = +1$ or -1 and $f_2(\omega) = +1$ or -1 .
- Plot the impedance (magnitude and phase) versus frequency.

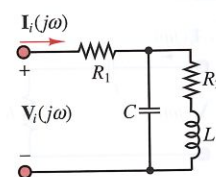


Figure P6.8

6.9 In the circuit of Figure P6.9:

$$R_1 = 1.3 \text{ k}\Omega \quad R_2 = 1.9 \text{ k}\Omega$$

$$C = 0.5182 \mu\text{F}$$

Determine:

- How the voltage frequency response function

$$H_v(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

behaves at extremes of high and low frequencies.

- An expression for the voltage frequency response function and show that it can be manipulated into the form

$$H_v(j\omega) = \frac{H_o}{1 + jf(\omega)}$$

where

$$H_o = \frac{R_2}{R_1 + R_2} \quad f(\omega) = \frac{\omega R_1 R_2 C}{R_1 + R_2}$$

- The frequency at which $f(\omega) = 1$ and the value of H_o in decibels.

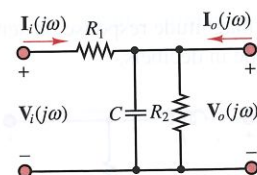


Figure P6.9

6.10 The circuit shown in Figure P6.10 is a second-order circuit because it has two reactive components (L and C). A complete solution will not be attempted. However, determine:

- The behavior of the voltage frequency response at extremely high and low frequencies.
- The output voltage V_o if the input voltage has a frequency where:

$$V_i = 7.07 \angle \frac{\pi}{4} \text{ V} \quad R_1 = 2.2 \text{ k}\Omega$$

$$R_2 = 3.8 \text{ k}\Omega \quad X_c = 5 \text{ k}\Omega \quad X_L = 1.25 \text{ k}\Omega$$

- The output voltage if the frequency of the input voltage doubles so that

$$X_c = 2.5 \text{ k}\Omega \quad X_L = 2.5 \text{ k}\Omega$$

- The output voltage if the frequency of the input voltage again doubles so that

$$X_c = 1.25 \text{ k}\Omega \quad X_L = 5 \text{ k}\Omega$$

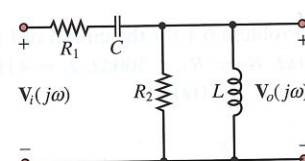


Figure P6.10

6.11 In the circuit shown in Figure P6.11, determine the frequency response function in the form:

$$H_v(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{H_{vo}}{1 \pm jf(\omega)}$$

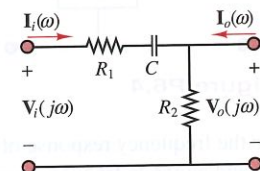


Figure P6.11

6.12 The circuit shown in Figure P6.12 has

$$R_1 = 100 \Omega \quad R_o = 100 \Omega$$

$$R_2 = 50 \Omega \quad C = 80 \text{ nF}$$

Determine the frequency response $V_o(j\omega)/V_i(j\omega)$.

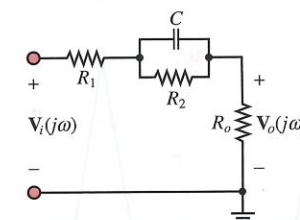


Figure P6.12

6.13

- Determine the frequency response $V_{out}(j\omega)/V_{in}(j\omega)$ for the circuit of Figure P6.13.
- Plot the magnitude and phase of the circuit for frequencies between 1 and 100 rad/s on graph paper, with a linear scale for frequency.
- Repeat part b, using semilog paper. (Place the frequency on the logarithmic axis.)
- Plot the magnitude response on semilog paper with magnitude in dB.

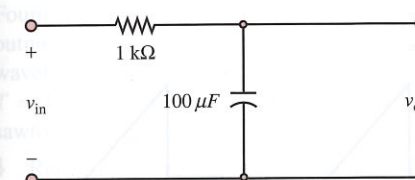


Figure P6.13

6.14 Consider the circuit shown in Figure P6.14.

- Sketch the amplitude response of $Y = I/V_S$.
- Sketch the amplitude response of V_1/V_S .
- Sketch the amplitude response of V_2/V_S .

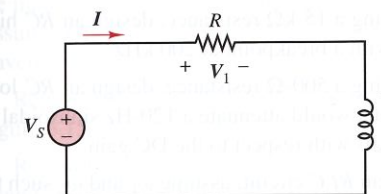


Figure P6.14

Section 6.2: Fourier Analysis

6.15 Use trigonometric identities to show that the equalities in equations 6.16 and 6.17 hold.

6.16 Derive a general expression for the Fourier series coefficients of the square wave of Figure 6.15(a) in the text.

6.17 Compute the Fourier series coefficient of the periodic function shown in Figure P6.17 and defined as:

$$x(t) = \begin{cases} 0 & 0 \leq t \leq \frac{T}{3} \\ A & \frac{T}{3} \leq t \leq T \end{cases}$$

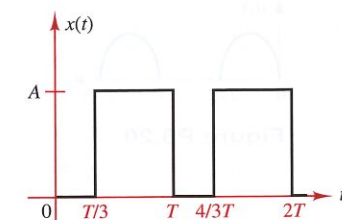


Figure P6.17

6.18 Compute the Fourier series coefficient of the periodic function shown in Figure P6.18 and defined as

$$x(t) = \begin{cases} \cos\left(\frac{2\pi}{T}t\right) & -\frac{T}{4} \leq t \leq \frac{T}{4} \\ 0 & \text{else} \end{cases}$$

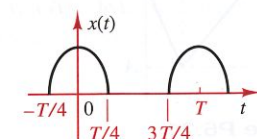


Figure P6.18

6.19 Compute the Fourier series expansion of the function shown in Figure P6.19, and express it in sine-cosine (a_n , b_n coefficients) form.

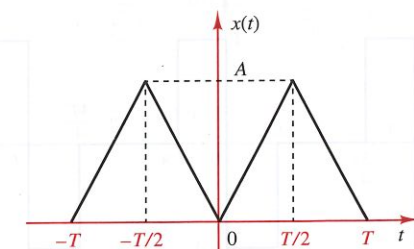


Figure P6.19

- 6.20** Compute the Fourier series expansion of the function shown in Figure P6.20, and express it in sine-cosine (a_n , b_n coefficients) form.

$$x(t) = \begin{cases} \sin\left(\frac{2\pi}{T}t\right) & 0 \leq t < \frac{T}{2} \\ 0 & \frac{T}{2} \leq t < T \end{cases}$$

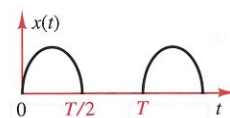


Figure P6.20

- 6.21** Write an expression for the signal shown in Figure P6.21, and derive a complete expression for its Fourier series.

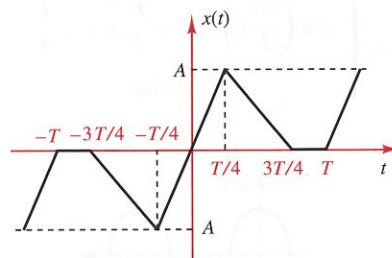


Figure P6.21

- 6.22** Write an expression for the signal shown in Figure P6.22 and derive its Fourier series.

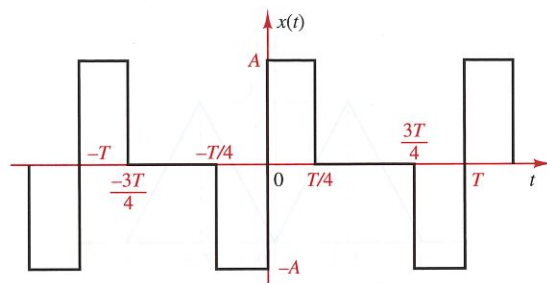


Figure P6.22

- 6.23** Find the Fourier series for the periodic function shown in Figure P6.23. Determine integral expressions for the Fourier coefficients.

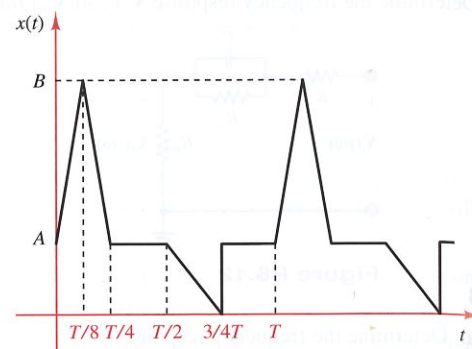


Figure P6.23

- 6.24** Find the Fourier series for the periodic function shown in Figure P6.24. Determine integral expressions for the Fourier coefficients.



Figure P6.24

Section 6.3: Filters

- 6.25** Using a 15-k Ω resistance, design an RC high-pass filter with a breakpoint at 200 kHz.
- 6.26** Using a 500- Ω resistance, design an RC low-pass filter that would attenuate a 120-Hz sinusoidal voltage by 20 dB with respect to the DC gain.
- 6.27** In an RLC circuit, assume ω_1 and ω_2 such that $I(j\omega_1) = I(j\omega_2) = I_{\max}/\sqrt{2}$ and $\Delta\omega$ such that $\Delta\omega = \omega_2 - \omega_1$. In other words, $\Delta\omega$ is the width of the current curve where the current has fallen to $1/\sqrt{2} = 0.707$ of its maximum value at the resonance frequency. At these frequencies, the power dissipated in a resistance becomes one-half of the dissipated

power at the resonance frequency (they are called the half-power points). In an RLC circuit with a high quality factor, show that $Q = \omega_0/\Delta\omega$.

- 6.28** In an RLC circuit with a high quality factor:

- Show that the impedance at the resonance frequency becomes a value of Q times the inductive resistance at the resonance frequency.
- Determine the impedance at the resonance frequency, assuming $L = 280$ mH, $C = 0.1$ μ F, $R = 25$ Ω .

- 6.29** At what frequency is the phase shift introduced by the circuit of Example 6.7 equal to -10° .

- 6.30** At what frequency is the output of the circuit of Example 6.7 attenuated by 10 percent (that is, $V_o = 0.9V_s$).

- 6.31** At what frequency is the output of the circuit of Example 6.11 attenuated by 10 percent (that is, $V_o = 0.9V_s$).

- 6.32** At what frequency is the phase shift introduced by the circuit of Example 6.11 equal to 20° .

- 6.33** Consider that the filter shown in Figure P6.1 is excited by a sawtooth waveform and that we are only interested in the response to the first two Fourier components of the waveform. Determine the output of the filter, and plot the input and output waveforms on the same graph. Assume the period $T = 10$ μ s and the peak amplitude $A = 1$ for the sawtooth waveform.

- 6.34** Repeat Problem 6.33 with the square wave of Figure 6.15(a) as an input.

- 6.35** Repeat Problem 6.33 for the pulse train of Example 6.4 as an input.

- 6.36** Consider that the circuit shown in Figure P6.2 is excited by the sawtooth waveform of Example 6.3 and that we are only interested in the response to the first three Fourier components of the waveform. Determine the output of the filter, and plot the input and output waveforms on the same graph. Assume $T = 0.5$ s and $A = 2$ for the sawtooth waveform.

- 6.37** Repeat Problem 6.36 with the square wave of Figure 6.15(a) as an input.

- 6.38** Repeat Problem 6.36 with the pulse train of Example 6.4 as an input.

- 6.39** Consider that the filter shown in Figure P6.3 is excited by the sawtooth waveform of Example 6.3 and that we are only interested in the response to the first four Fourier components of the waveform. Determine the output of the filter, and plot the input and output

waveforms on the same graph. Assume $T = 0.1$ s and $A = 1$ for the sawtooth waveform.

- 6.40** Repeat Problem 6.39 with the square wave of Figure 6.15(a) as an input.

- 6.41** Repeat Problem 6.39 with the pulse train of Example 6.4 as an input.

- 6.42** Consider that the filter shown in Figure P6.4 is excited by a sawtooth waveform of Example 6.3 and that we are only interested in the response to the first two Fourier components of the waveform. Determine the output of the filter, and plot the input and output waveforms on the same graph. Assume $T = 50$ ms and $A = 2$ for the sawtooth waveform.

- 6.43** Repeat Problem 6.42 for $T = 0.5$ s and 5 ms, and compare the results with $T = 50$ ms.

- 6.44** Repeat Problem 6.42 for the square wave of Figure 6.15(a).

- 6.45** Repeat Problem 6.42 with the pulse train of Example 6.4 as an input.

- 6.46** Consider that the filter shown in Figure P6.5 is excited by the sawtooth waveform of Example 6.3 and that we are only interested in the response to the first three Fourier components of the waveform. Determine the output of the filter, and plot the input and output waveforms on the same graph. Assume $T = 5$ s and $A = 1$ for the sawtooth waveform.

- 6.47** Repeat Problem 6.46 for $T = 50$ s, and compare the results.

- 6.48** Repeat Problem 6.46 with the square wave of Figure 6.15(a) as an input.

- 6.49** Repeat Problem 6.46 with the pulse train of Example 6.4 as an input.

- 6.50** Consider the circuit shown in Figure P6.50. Determine the resonant frequency and the bandwidth for the circuit.

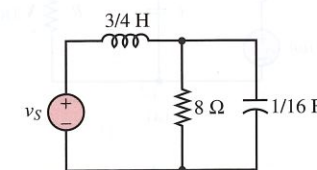
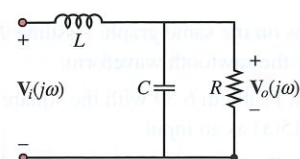
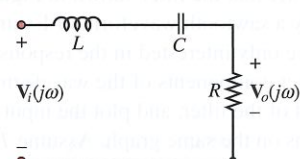


Figure P6.50

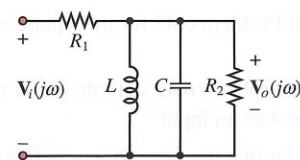
- 6.51** Are the filters shown in Figure P6.51 low-pass, high-pass, bandpass, or bandstop (notch) filters?



(a)



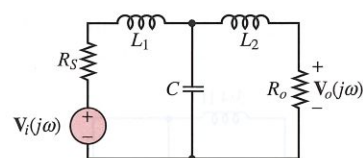
(b)



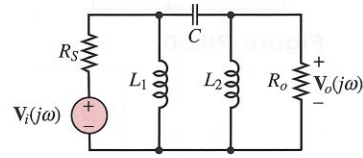
(c)

Figure P6.51

6.52 Determine if each of the circuits shown in Figure P6.52 is a low-pass, high-pass, bandpass, or bandstop (notch) filter.

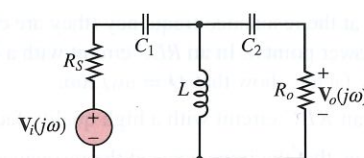


(a)

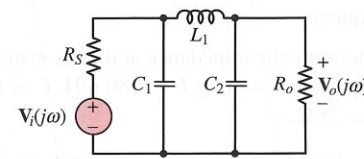


(b)

Figure P6.52



(c)



(d)

Figure P6.52

6.53 For the filter circuit shown in Figure P6.53:

- Determine if this is a low-pass, high-pass, bandpass, or bandstop filter.
- Determine the frequency response $V_o(j\omega)/V_i(j\omega)$ assuming $L = 10 \text{ mH}$, $C = 1 \text{ nF}$, $R_1 = 50 \Omega$, $R_2 = 2.5 \text{ k}\Omega$.

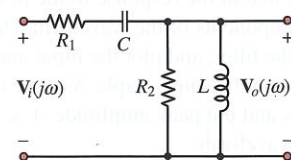


Figure P6.53

6.54 In the filter circuit shown in Figure P6.54: $L = 10 \text{ H}$, $C = 1 \text{ nH}$, $R_s = 20 \Omega$, $R_c = 100 \Omega$, $R_o = 5 \text{ k}\Omega$. Determine the frequency response $V_o(j\omega)/V_i(j\omega)$. What type of filter does this frequency response represent?

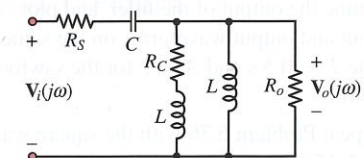


Figure P6.54

6.55 In the filter circuit shown in Figure P6.54: $L = 0.1 \text{ mH}$, $C = 8 \text{ nH}$, $R_s = 300 \Omega$, $R_c = 10 \Omega$, $R_o = 500 \Omega$. Determine the frequency response $V_o(j\omega)/V_i(j\omega)$. What type of filter does this frequency response represent?

6.56 In the filter circuit shown in Figure P6.56:

$$R_s = 5 \text{ k}\Omega \quad C = 56 \text{ nF}$$

$$R_o = 100 \text{ k}\Omega \quad L = 9 \text{ }\mu\text{H}$$

Determine:

- The voltage frequency response

$$H_v(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

- The resonant frequency.
- The half-power frequencies.
- The bandwidth and Q .

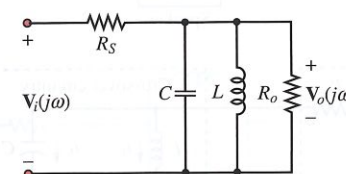


Figure P6.56

6.57 In the filter circuit shown in Figure P6.56:

$$R_s = 5 \text{ k}\Omega \quad C = 0.5 \text{ nF}$$

$$R_o = 100 \text{ k}\Omega \quad L = 1 \text{ mH}$$

Determine:

- The voltage frequency response

$$H_v(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

- The resonant frequency.
- The half-power frequencies.
- The bandwidth and Q .

6.58 In the filter circuit shown in Figure P6.58:

$$R_s = 500 \Omega \quad R_o = 5 \text{ k}\Omega$$

$$R_c = 4 \text{ k}\Omega \quad L = 1 \text{ mH}$$

$$C = 5 \text{ pF}$$

Determine the frequency response $H(j\omega)$, where:

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

What type of filter does this frequency response represent?

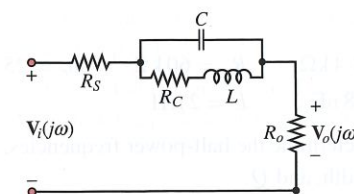


Figure P6.58

6.59 In the notch filter circuit shown in Figure P6.59, derive the voltage frequency response $H(j\omega)$ in standard form, where:

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

Assume:

$$R_s = 500 \Omega \quad R_o = 5 \text{ k}\Omega$$

$$C = 5 \text{ pF} \quad L = 1 \text{ mH}$$

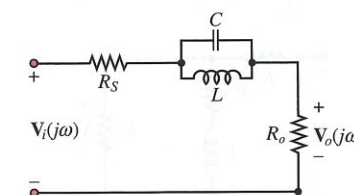


Figure P6.59

6.60 In the notch filter circuit shown in Figure P6.59, derive the voltage frequency response $H(j\omega)$ in standard form, where:

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

Assume:

$$R_s = 500 \Omega \quad R_o = 5 \text{ k}\Omega$$

$$\omega_r = 12.1278 \text{ Mrad/s} \quad C = 68 \text{ nF}$$

$$L = 0.1 \text{ }\mu\text{H}$$

Also, determine the half-power frequencies, bandwidth, and Q .

6.61 In the notch filter circuit shown in Figure P6.59, derive the voltage frequency response $H(j\omega)$ in standard form, where:

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)}$$

Assume:

$$R_S = 4.4 \text{ k}\Omega \quad R_o = 60 \text{ k}\Omega \quad \omega_r = 25 \text{ Mrad/s}$$

$$C = 0.8 \text{ nF} \quad L = 2 \text{ }\mu\text{H}$$

Also, determine the half-power frequencies, bandwidth, and Q .

6.62 In the bandstop (notch) filter shown in Figure P6.62:

$$L = 0.4 \text{ mH} \quad R_c = 100 \text{ }\Omega$$

$$C = 1 \text{ pF} \quad R_S = R_o = 3.8 \text{ k}\Omega$$

Determine:

a. An expression for the voltage frequency response:

$$\mathbf{H}_v(j\omega) = \frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)} = H_o \frac{1 + jf_1(\omega)}{1 + jf_2(\omega)}$$

- b. The magnitude of the frequency response at very high and very low frequencies and at the resonant frequency.
- c. The magnitude of the frequency response at the resonant frequency.
- d. The resonant and half-power frequencies.

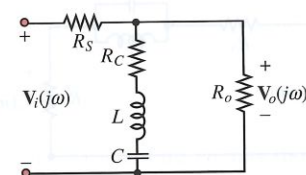


Figure P6.62

6.63 In the filter circuit shown in Figure P6.56, assume:

$$R_S = 5 \text{ k}\Omega \quad C = 5 \text{ nF}$$

$$R_o = 50 \text{ k}\Omega \quad L = 2 \text{ mH}$$

Determine:

a. An expression for the voltage frequency response function

$$\mathbf{H}_v(j\omega) = \frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)}$$

- b. The resonant frequency.
- c. The half-power frequencies.
- d. The bandwidth and Q .

6.64 Many stereo speakers are two-way speaker systems; that is, they have a woofer for low-frequency sounds and a tweeter for high-frequency sounds. To get the proper separation of frequencies going to the woofer and to the tweeter, crossover circuitry is used. A crossover circuit is effectively a bandpass,

high-pass, or low-pass filter. The system model is shown in Figure P6.64. The function of the crossover circuitry is to channel frequencies below a given crossover frequency, f_c , into the woofer and frequencies higher than f_c into the tweeter. Assume an ideal amplifier such that $R_S = 0$ and that the desired crossover frequency is 1,200 Hz. Find C and L when $R_1 = R_2 = 8 \text{ }\Omega$. [Hint: Set the break frequency of the network seen by the amplifier equal to the desired crossover frequency.]

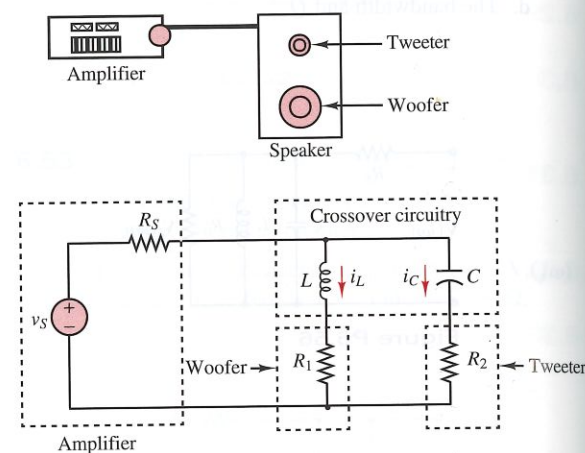


Figure P6.64

Section 6.4: Bode Plots

6.65 Determine the frequency response $\mathbf{V}_{out}(\omega)/\mathbf{V}_S(\omega)$ for the network in Figure P6.65. Generate the Bode magnitude and phase plots when $R_S = R_o = 5 \text{ k}\Omega$, $L = 10 \text{ }\mu\text{H}$, and $C = 0.1 \text{ }\mu\text{F}$.

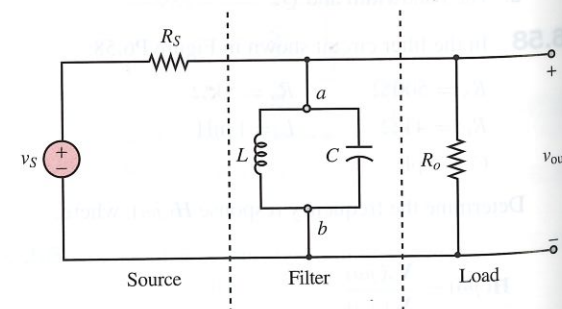


Figure P6.65

6.66 Refer to Problem 6.64 but assume that $L = 2 \text{ mH}$, $C = 125 \text{ }\mu\text{F}$, and $R_S = R_1 = R_2 = 4 \text{ }\Omega$ in Figure P6.64.

- a. Determine the impedance seen by the amplifier as a function of frequency. At what frequency is maximum power transferred by the amplifier?
- b. Generate the Bode magnitude and phase plots of the currents through the woofer and tweeter.

6.67 For the notch filter shown in Figure P6.67 assume that $R_S = R_o = 500 \text{ }\Omega$, $L = 10 \text{ mH}$, and $C = 0.1 \text{ }\mu\text{F}$.

- a. Determine the frequency response $\mathbf{V}_{out}(j\omega)/\mathbf{V}_S(j\omega)$.
- b. Generate the associated Bode magnitude and phase plots.

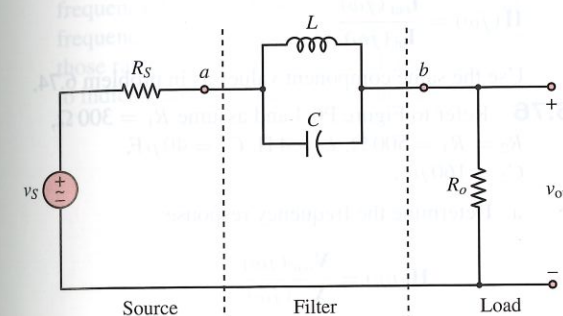


Figure P6.67

6.68 It is very common to see interference caused by power lines, at a frequency of 60 Hz. This problem outlines the design of the notch filter shown in Figure P6.68 to reject a band of frequencies around 60 Hz.

- a. Determine the impedance $\mathbf{Z}_{ab}(j\omega)$ between nodes a and b for the filter of Figure P6.68. r_L represents the resistance of a practical inductor.
- b. For what value of C will the center frequency of $\mathbf{Z}_{ab}(j\omega)$ equal 60 Hz when $L = 100 \text{ mH}$ and $r_L = 5 \text{ }\Omega$?
- c. Would the "sharpness," or selectivity, of the filter increase or decrease if r_L were increased?
- d. Assume that the filter is used to eliminate the 60-Hz noise from a 1 kHz sine wave. Evaluate the frequency response $\mathbf{V}_o/\mathbf{V}_{in}(j\omega)$ at both frequencies when:

$$v_g(t) = \sin(2\pi 1,000 t) \text{ V} \quad r_g = 50 \text{ }\Omega$$

$$v_n(t) = 3 \sin(2\pi 60 t) \quad R_o = 300 \text{ }\Omega$$

Assume $L = 100 \text{ mH}$ and $r_L = 5 \text{ }\Omega$. Use the value of C found in part b.

- e. Generate the Bode magnitude and phase plots for $\mathbf{V}_o/\mathbf{V}_{in}$. Mark the plots at 60 Hz and 1,000 Hz.

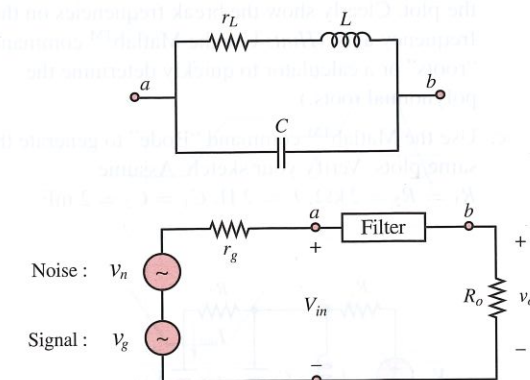


Figure P6.68

6.69 The circuit of Figure P6.69 is representative of an amplifier-speaker connection. The crossover filter allows low-frequency signals to pass to the woofer. The filter's topography is known as a π network.

- a. Find the frequency response $\mathbf{V}_o(j\omega)/\mathbf{V}_S(j\omega)$.
- b. If $C_1 = C_2 = C$, $R_S = R_o = 600 \text{ }\Omega$, and $1/\sqrt{LC} = R/L = 1/RC = 2,000\pi$, generate the Bode magnitude and phase plots in the range $100 \text{ Hz} \leq f \leq 10 \text{ kHz}$.

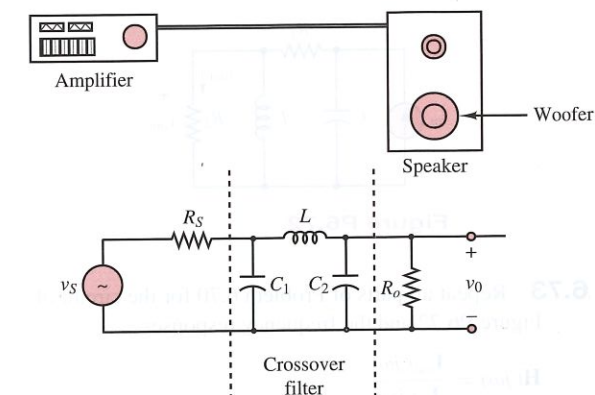


Figure P6.69

6.70 For the circuit shown in Figure P6.70:

- a. Determine the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{V}_{\text{out}}(j\omega)}{\mathbf{V}_{\text{in}}(j\omega)}$$

- b. Sketch, by hand, the associated Bode magnitude and phase plots. List all the steps in constructing the plot. Clearly show the break frequencies on the frequency axis. (Hint: Use the Matlab™ command “roots” or a calculator to quickly determine the polynomial roots.)
- c. Use the Matlab™ command “Bode” to generate the same plots. Verify your sketch. Assume $R_1 = R_2 = 2 \text{ k}\Omega$, $L = 2 \text{ H}$, $C_1 = C_2 = 2 \text{ mF}$.

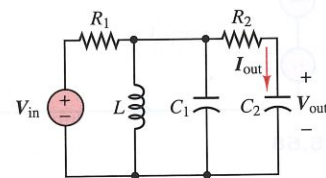


Figure P6.70

6.71 Repeat all parts of Problem 6.70 for the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{I}_{\text{out}}(j\omega)}{\mathbf{V}_{\text{in}}(j\omega)}$$

Use the same component values as in Problem 6.70.

6.72 Repeat all parts of Problem 6.70 for the circuit of Figure P6.72 and the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{V}_{\text{out}}(j\omega)}{\mathbf{I}_{\text{in}}(j\omega)}$$

Let $R_1 = R_2 = 1 \text{ k}\Omega$, $C = 1 \text{ }\mu\text{F}$, $L = 1 \text{ H}$.

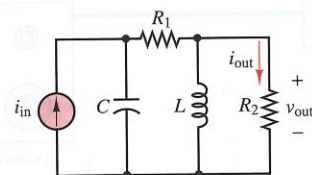


Figure P6.72

6.73 Repeat all parts of Problem 6.70 for the circuit of Figure P6.72 and the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{I}_{\text{out}}(j\omega)}{\mathbf{I}_{\text{in}}(j\omega)}$$

Use the same values as in Problem 6.72.

6.74 For the circuit of Figure P6.74 determine the frequency response $\mathbf{H}(j\omega) = \mathbf{V}_{\text{out}}/\mathbf{I}_{\text{in}}$. Assume $R_1 = R_2 = 2 \text{ k}\Omega$, $C_1 = C_2 = 1 \text{ mF}$.

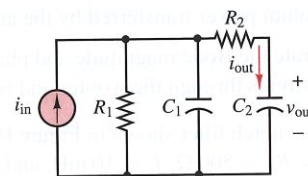


Figure P6.74

6.75 Repeat all parts of Problem 6.70 for the circuit of Figure P6.74 and the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{I}_{\text{out}}(j\omega)}{\mathbf{I}_{\text{in}}(j\omega)}$$

Use the same component values as in problem 6.74.

6.76 Refer to Figure P6.4 and assume $R_1 = 300 \text{ }\Omega$, $R_2 = R_3 = 500 \text{ }\Omega$, $L = 4 \text{ H}$, $C_1 = 40 \text{ }\mu\text{F}$, $C_2 = 160 \text{ }\mu\text{F}$.

- a. Determine the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{V}_{\text{out}}(j\omega)}{\mathbf{V}_{\text{in}}(j\omega)}$$

- b. Sketch, by hand, the associated Bode magnitude and phase plots. List all the steps in constructing the plot. Clearly show the break frequencies on the frequency axis. (Hint: Use the Matlab™ command “roots” or a calculator to quickly determine the polynomial roots.)
- c. Use the Matlab™ command “Bode” to generate the same plots. Verify your sketch.

6.77 Refer to Figure P6.4 and the parameter values listed in Problem 6.76.

- a. Determine for the frequency response:

$$\mathbf{H}(j\omega) = \frac{\mathbf{V}_C(j\omega)}{\mathbf{V}_{\text{in}}(j\omega)}$$

- b. Repeat parts b and c of Problem 6.76 for this frequency response.

6.78 Refer to Figure P6.5 and repeat the instructions of parts b and c of Problem 6.76. Assume $R_1 = 20 \text{ k}\Omega$, $R_2 = 100 \text{ k}\Omega$, $L = 1 \text{ H}$, $C = 100 \text{ }\mu\text{F}$.

6.79 Assume in a certain frequency range that the ratio of output amplitude to input amplitude is proportional to $1/\omega^3$. What is the slope of the Bode magnitude plot in this frequency range, expressed in dB/decade?

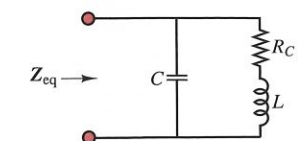
6.80 Assume that the amplitude of an output voltage depends on frequency according to:

$$\mathbf{V}(j\omega) = \frac{A\omega + B}{\sqrt{C + D\omega^2}}$$

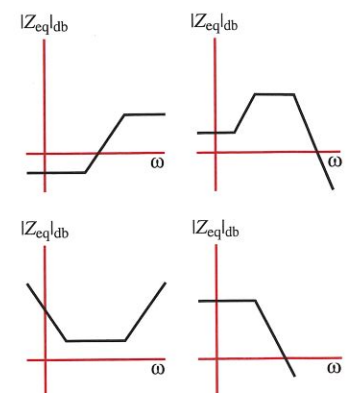
Find:

- a. The break frequency.
- b. The slope (in dB/decade) of the Bode magnitude plot above the break frequency.
- c. The slope (in dB/decade) of the Bode plot below the break frequency.
- d. The high-frequency limit of $\mathbf{V}(j\omega)$.

6.81 Determine the equivalent impedance $\mathbf{Z}_{\text{eq}}$ in standard form as defined in Figure P6.81(a). Choose the Bode plot from Figure P6.81(b) that best describes the behavior of the impedance as a function of frequency. Describe how to find the resonant and cutoff frequencies, and the magnitude of the impedance for those ranges where it is constant. Label the Bode plot to indicate which feature you are discussing.



(a)



(b)

Figure P6.81